The Master's Thesis

Career Path Guidance Model with Multi-Label Classification

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Master's of Science in Big Data Analytics

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The Graduate Program

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Abstract

The alignment of a student's academic capabilities with an appropriate career trajectory is a foundational challenge in modern educational systems. In Rwanda, students transition from Ordinary Level (Senior 3) to Advanced Level based on their national examination performance, a critical juncture that heavily dictates their future career options, particularly within the highly demanding health sector. Traditional guidance systems rely heavily on manual counseling, which is often subjective, constrained by cognitive biases, and incapable of processing large-scale historical performance data. Furthermore, early algorithmic approaches to career guidance typically frame the problem as a mutually exclusive, single-label classification task, functioning as "black boxes" that fail to explain the mathematical reasoning behind their recommendations.

This thesis develops and evaluates an Explainable Multi-Label Machine Learning Model designed to provide transparent, multi-pathway educational and career guidance. Utilizing a comprehensive longitudinal dataset of Senior 3 national examination results spanning from 2017 to 2025, the study focuses on core STEM subjects (Biology, Chemistry, Mathematics, and Physics). The methodology employs the Classifier Chains technique to account for the inherent correlations between related health sector careers (e.g., Medicine and Nursing). Four distinct base algorithms, Decision Trees, Random Forests, Extreme Gradient Boosting (XGBoost), and Multi-Layer Perceptrons (MLP), are evaluated based on Subset Accuracy, Hamming Loss, and Micro F1-Scores.

Crucially, the system moves beyond mere prediction by integrating an Explainable AI (XAI) layer. It utilizes feature importance mapping to provide global interpretability and introduces the mathematical framework for Counterfactual Explanations to offer local, actionable advice (e.g., identifying the minimum required improvement in specific subjects to change a rejected recommendation to an accepted one). The results demonstrate that ensemble methods, combined with the Classifier Chains mechanism, significantly outperform independent binary classifiers in capturing complex, non-linear relationships in student data while preserving high interpretability.

This research establishes a robust, mathematically sound, and scalable framework that not only predicts but explicitly explains career recommendations, laying the theoretical groundwork for future integration with longitudinal tracking, continuous adaptation, and dynamic labor market data.

Keywords: Educational Data Mining, Multi-Label Classification, Classifier Chains, Explainable AI, XGBoost, Random Forest, Counterfactual Explanations, Career Guidance.

Introduction

The process of choosing a career is undeniably one of the most critical and defining decisions a student will make in their lifetime. This single choice not only dictates their higher education trajectory but also establishes the foundation for their socioeconomic mobility, personal fulfillment, and contribution to national development. However, despite the gravity of this decision, many students make this choice based on highly constrained and often skewed information. In developing nations particularly, peer pressure, societal expectations, and parental demands frequently overshadow a clear, objective understanding of the student's own academic strengths, weaknesses, and intrinsic aptitudes. This misalignment is particularly detrimental when attempting to guide students toward the health sector, an arena that demands absolute precision, resilience, and specific, deep foundational knowledge in core science subjects such as Biology, Chemistry, Mathematics, and Physics.

To address this systemic challenge, researchers and educators have increasingly turned toward intelligent guidance systems. These systems leverage the vast amounts of historical data generated by educational institutions to analyze a student's past academic performance and subsequently recommend highly suitable educational pathways and careers. However, traditional systems deployed in educational data mining have often treated this complex, multifaceted challenge as a simple, mutually exclusive classification problem. They provide a singular, absolute recommendation (e.g., "You should become a Doctor") without exploring alternative viable paths and, crucially, without mathematically or logically explaining *why* the recommendation was made.

This thesis introduces a comprehensive conceptual framework designed to revolutionize this approach. The framework models the student journey not as a single point of decision, but as a continuous, intelligent, and interconnected pathway:

$$\boxed{\text{Academic Performance (Input)} \rightarrow \text{Educational Pathway} \rightarrow \text{Career Pathway (Output)}}$$

Specifically, this study develops an Explainable Multi-Label (Tsoumakas & Katakis, 2007) Machine Learning Model designed explicitly to assist students in Rwanda transitioning from their Ordinary Level (Senior 3 or S3) to Advanced Level (S6) and, ultimately, into highly specialized health sector careers. The system proposed herein integrates multiple layers of artificial intelligence, including prediction, multi-label (Tsoumakas & Katakis, 2007) recommendation, and explicit mathematical explanation, moving far beyond simple black-box algorithmic models to provide actionable, ranked, and logically transparent guidance that students and educators can actively trust.

Background of the Study

In the Rwandan education system, the academic journey is sharply bifurcated at critical assessment junctures. Students complete their Ordinary Level (S3) education by taking rigorous national examinations administered by the National Examination and School Inspection Authority (NESA). The results from these national exams, particularly in the core STEM subjects (Science, Technology, Engineering, and Mathematics), serve as the strongest, most objective indicators of a student's cognitive aptitude and academic preparedness. The transition from S3 to Advanced Level (S6) is not merely a change in grade; it is a definitive filtering mechanism where students are channeled into highly specialized subject combinations (such as PCB: Physics-Chemistry-Biology, or MCB: Mathematics-Chemistry-Biology). These combinations strictly dictate their eligibility for future university degrees and directly constrain their options in the labor market. If a student with a high aptitude for healthcare mistakenly chooses a combination lacking Biology or Chemistry due to poor guidance, their path to becoming a medical professional is essentially severed.

The health sector represents a critical, high-priority area for Rwanda's national development. Under the framework of Rwanda Vision 2050 and the National Strategy for Transformation (NST1), the government has explicitly prioritized the development of a knowledge-based economy and the establishment of world-class healthcare infrastructure. To achieve these ambitious macroeconomic and social goals, the nation requires a steady, high-quality, and predictable pipeline of capable professionals, including specialized physicians, advanced practice nurses, biomedical engineers, pharmacists, and public health administrators. Consequently, guiding the right students, those possessing the precise mathematical, analytical, and scientific aptitudes required to survive rigorous medical training, into the appropriate educational pathways is not just an individual necessity, but a macroeconomic imperative that directly impacts national mortality rates and healthcare efficiency.

Historically, this critical guidance has been conducted manually by educators, school administrators, and career counselors. However, human-led guidance is inherently limited in scope and accuracy. It is highly susceptible to cognitive biases, constrained by the very limited time available per student, and mathematically incapable of processing the massive, multi-dimensional historical datasets required to recognize subtle, complex patterns of long-term academic success and failure. For instance, a human counselor cannot instantly calculate the precise probability of a student succeeding in Medical School given an 82% in Chemistry, a 74% in Physics, and a 91% in Biology compared against the historical outcomes of ten thousand similar students.

Machine learning (ML) offers a profound, scalable solution by autonomously identifying these complex, non-linear patterns in large historical datasets. Yet, despite the immense promise of Artificial Intelligence, typical ML approaches utilized in educational data mining and career prediction harbor significant methodological and theoretical limitations that prevent widespread adoption:

1. **Single-Label Constraint:** Traditional models almost exclusively predict only a single career outcome. They frame the complex human capability matrix as a mutually exclusive choice (e.g., classifying a student strictly as either a Doctor OR a Nurse). This ignores the

fundamental reality of human aptitude, that a student's unique skills might make them highly suitable for multiple related health sector careers simultaneously.

2. **The Black-Box Dilemma:** Advanced models, particularly deep neural networks and complex ensembles, operate as opaque "black boxes." They provide a final prediction without any transparent mathematical or logical reasoning. In a high-stakes domain like educational tracking, providing a prediction without an explanation destroys user trust. If a student is told by an algorithm that they are statistically unlikely to succeed in Medicine, they must mathematically understand exactly which specific academic deficiencies led to that conclusion in order to accept it or attempt to improve.

3. **Static Thresholding:** Many legacy guidance systems rely on hard-coded static thresholds (e.g., "Must have $> 80\%$ in Biology to study Medicine"), failing to account for the dynamic interactions and covariance between different STEM subjects that a machine learning model inherently captures.

Problem Statement

Despite the modern availability of vast, rich amounts of educational data (such as the comprehensive S3 national exam datasets collected from 2017 to 2025 across Rwanda), there is a severe lack of advanced, multi-label (Tsoumakas & Katakis, 2007), and explainable machine learning models utilized for career path guidance.

Existing predictive models generally treat career guidance as a mutually exclusive choice. By framing the algorithm as a standard multi-class problem, they attempt to classify a student strictly as a future "Doctor" OR a "Nurse" OR a "Pharmacist." This approach is fundamentally flawed because it forces the algorithm to suppress secondary, highly viable options, thereby failing to output ranked probabilities across multiple pathways that the student might excel in.

Furthermore, these traditional systems completely lack the critical intelligence layer of explainability. When a student is strongly recommended to pursue Pharmacy, current predictive systems cannot mathematically, visually, or logically explain to the student or their parents which specific exam scores contributed most heavily to that specific recommendation. They cannot answer the "Why?"

This study rigorously addresses these empirical and theoretical gaps by implementing a multi-label (Tsoumakas & Katakis, 2007) classification approach structurally combined with cutting-edge Explainable AI (XAI) techniques. The result is a system that not only predicts multiple career pathways simultaneously, acknowledging the multifaceted nature of student capability, but also explicitly explains the mathematical reasoning driving those recommendations.

Objective of the Study

General Objective:

To research, develop, and evaluate an explainable, multi-label (Tsoumakas & Katakis, 2007) machine learning model capable of accurately predicting educational and career pathways in the health sector based strictly on Rwandan S3 academic performance data.

Specific Objectives:

- To fundamentally restructure career prediction from a single-label to a multi-label (Tsoumakas & Katakis, 2007) classification paradigm by implementing the Classifier Chains (Read et al., 2011) mechanism.
- To train, tune, and comparatively evaluate multiple base machine learning algorithms, specifically Extreme Gradient Boosting (XGBoost), Random Forest (Breiman, 2001), Decision Trees, and Multi-Layer Perceptrons (MLP).
- To generate ranked, probabilistic outputs for multiple career outcomes rather than a single, absolute, and limiting prediction.

- To integrate a robust explainability layer utilizing feature importance mapping to explicitly identify and quantify which academic subjects drive specific career recommendations.
- To explore and formulate the fundamental mathematics of counterfactual (Wachter et al., 2017) explanations, providing a theoretical mechanism to show students exactly what academic score changes would alter their career recommendations.

Research Questions

To achieve the aforementioned objectives, this study seeks to systematically answer the following research questions:

1. How effectively can a multi-label (Tsoumakas & Katakis, 2007) classification approach, specifically Classifier Chains (Read et al., 2011), model the non-mutually exclusive, highly correlated nature of health sector career pathways compared to traditional single-label methods?
2. How do advanced tree-based ensemble models (XGBoost, Random Forest) compare to foundational models (Decision Tree) and deep learning models (MLP) in terms of Subset Accuracy and Hamming Loss when predicting multi-label (Tsoumakas & Katakis, 2007) career paths based on exam scores?
3. In what ways can Explainable AI (XAI) techniques mathematically clarify the complex, non-linear relationships between specific S3 subject scores and the resulting multi-label (Tsoumakas & Katakis, 2007) career recommendations?

Hypotheses

Based on the theoretical framework and existing literature in the broader field of machine learning, this research proposes the following testable hypotheses:

- **H1 (Architectural Superiority):** Multi-label models combined with the Classifier Chains (Read et al., 2011) mechanism will yield significantly higher predictive subset accuracy and lower Hamming Loss for career recommendations than completely independent, unchained binary classifiers (Binary Relevance), because they capture the inherent label correlations between related health careers.
- **H2 (Algorithmic Dominance):** Ensemble tree-based methods, specifically XGBoost (Chen & Guestrin, 2016) and Random Forest (Breiman, 2001), will consistently outperform singular Decision Trees and basic Multi-Layer Perceptrons in capturing the specific complex, non-linear, and non-normalized relationships present in raw student grading data.
- **H3 (Interpretability Enhancement):** The structural integration of an explainability layer will successfully extract global feature importance, allowing for a clear, verifiable mathematical mapping between the input features (academic scores in Biology, Chemistry, Math, Physics) and the model's prediction output, thereby resolving the black-box dilemma.

Significance of the Study

The primary significance of this research lies in its paradigm shift away from basic, isolated algorithmic comparison and toward the creation of a comprehensive, human-centric, and highly intelligent educational framework. By explicitly prioritizing mathematical explainability alongside multi-label (Tsoumakas & Katakis, 2007) probability ranking, this study provides a highly practical, deployable tool that addresses the exact needs of various stakeholders in the Rwandan educational ecosystem.

Impact on Students and Parents

For students and parents, the system offers unprecedented transparent guidance, mitigating the anxiety and uncertainty that traditionally surround the S3 to S6 transition. If a student is recommended for a Bachelor of Pharmacy with a high probability of $P = 0.85$, the model's explainability component will mathematically clarify that this is largely driven by their exceptionally high scores in Chemistry and Mathematics, which successfully offset a slightly lower score in Biology. Conversely, if a student is rejected from a Medical track, the counterfactual (Wachter et al., 2017) engine provides actionable feedback (e.g., "An increase of 8% in Physics would have changed this recommendation"). This transparency empowers students to make highly informed, logical decisions about their advanced educational pathways rather than relying on guesswork, societal pressure, or subjective human opinions.

Impact on Educational Institutions and Counselors

For educators, headteachers, and school counselors, the model acts as a highly advanced decision-support tool. It has the computational power to process thousands of historical data points instantly, effectively multiplying the counselor's capacity. By automating the objective, data-driven baseline recommendations, counselors are freed from manual transcript analysis. This allows them to focus their limited time entirely on the human, psychological, and emotional aspects of guidance, such as discussing a student's financial constraints, personal passions, or family expectations, while confidently relying on the algorithm for the empirical academic baseline.

Impact on National Healthcare Infrastructure and Policy

At a macroeconomic level, this model clearly demonstrates how historical national examination data, often left dormant in government databases after graduation, can be effectively leveraged as a form of predictive intelligence. By optimizing the pipeline of talent entering the critical health sector, this system ensures that university medical and nursing programs receive candidates who possess the precise, verified cognitive aptitudes required to succeed. This reduces university dropout rates, increases the quality of graduating healthcare professionals, and directly supports the long-term healthcare infrastructure goals outlined in Rwanda's National Strategy for Transformation (NST1).

Scope of the Study

To maintain a rigorous, mathematically sound, and feasible focus suitable for a Master's thesis, it is crucial to clearly define the boundaries of this research within the much broader conceptual vision of AI in education. This thesis explicitly divides the grand vision into current implementations and future doctoral-level extensions.

Included in this Thesis (The Master's Scope):

- **Prediction & Recommendation:** The complete implementation of multi-label (Tsoumakas & Katakis, 2007) classification to output mathematically ranked probabilities for various health careers based on cross-sectional historical data.
- **Explanation:** The implementation of feature importance metrics to explain the global behavior of the models.
- **Counterfactual Formulation:** Establishing the strict mathematical foundations and theoretical optimization formulas for basic counterfactual (Wachter et al., 2017) explanations (e.g., $x^* = \arg\min_{x'} d(x, x')$ subject to $f(x') \neq f(x)$).
- **Algorithms Evaluated:** XGBoost (Chen & Guestrin, 2016), Random Forest (Breiman, 2001), Decision Tree, and MLP using the Classifier Chains (Read et al., 2011) methodology.

Excluded from this Thesis (Future Research / The PhD Vision):

- **Continuous Evaluation (Dynamic Adaptation):** The proposed system will be trained on static historical data. It will not dynamically update its mathematical weights in real-time as new students are assessed ($R_t = f(X_t)$). Achieving this requires an active, longitudinal feedback loop outside the current scope.

- **Longitudinal Modeling:** Tracking the exact same cohort of students over many years through university to verify their actual final career placement against the S3 prediction.
- **Labour-Market Integration:** Incorporating external, fluctuating economic data, salary trends, or real-time hospital job market demand into the recommendation logic.
- **Advanced Career Graphing:** Building sophisticated Graph Neural Networks (GNN) to explicitly map the complex, multi-node subject-combination-career relationships as a mathematical graph.

By strictly defining these boundaries, this thesis establishes a highly ambitious yet technically achievable framework: a rigorous current implementation that serves as the concrete, proven foundation for expansive future extensions.

Limitations and Assumptions of the Study

While this research aims to provide a robust, highly accurate predictive model, it is fundamentally constrained by several empirical limitations and underlying mathematical assumptions that must be explicitly acknowledged:

Assumptions

- **Cognitive Reflection in Grading:** A foundational assumption of this model is that a student’s performance on the S3 national examinations in STEM subjects is a direct, objective, and accurate reflection of their cognitive aptitude for those subjects. It assumes the grading mechanism is perfectly standardized and devoid of systemic errors.
- **Stationarity of the Education System:** The model operates under the mathematical assumption of stationarity, meaning it assumes that the difficulty of the national exams, the grading rubrics, and the academic requirements for university health programs remain relatively constant between 2017 and 2025.

Limitations

- **Exclusion of Socio-Economic Variables:** The current model relies exclusively on quantitative academic scores. It does not possess data regarding a student's socio-economic status, geographical location, psychological resilience, or financial capacity to afford medical school. These hidden variables (Z) introduce unmeasured variance into the real-world probability of a student actually achieving the recommended career.
- **Static Data Snapshot:** The model is trained on a static, cross-sectional snapshot of historical data. It cannot track an individual student's intellectual growth or decline between S3 (Ordinary Level) and S6 (Advanced Level).
- **Subject Scope Constraint:** To maintain high predictive accuracy specifically for the health sector, the model's feature space is strictly constrained to Biology, Chemistry, Mathematics, and Physics. It ignores humanities or language scores, which, while generally less critical for clinical medicine, still play a role in a student's overall academic profile.

Operational Definition of Terms

To ensure clarity throughout the technical discussions in the following chapters, key terms are defined operationally as follows:

- **Multi-Label Classification:** A machine learning paradigm where each input instance (a student) can be assigned multiple target labels simultaneously (e.g., a student is mathematically classified as suited for both Medicine and Nursing). This contrasts with multi-class classification, where only one label can be chosen.
- **Explainable AI (XAI):** A suite of mathematical methods and algorithms designed to ensure that the internal mechanics, feature weights, and final results of an artificial intelligence solution can be thoroughly understood, audited, and trusted by human operators.

- **Classifier Chains (CC):** An advanced meta-algorithm for multi-label (Tsoumakas & Katakis, 2007) classification that links several binary classifiers in a sequential chain. Each classifier in the chain includes the predictions of all previous classifiers as additional input features, thereby allowing the model to capture deep statistical correlations between labels.

- **Counterfactual Explanation:** A specific form of local explanation that mathematically describes the absolute smallest theoretical change to the input data (e.g., increasing a Mathematics score by 4 points) that would result in the algorithm flipping its output to a different prediction.

- **Ensemble Learning:** A machine learning technique that strategically combines several base models (such as hundreds of Decision Trees) in order to produce one highly optimal, generalized predictive model that reduces variance (as in Random Forests) or bias (as in XGBoost).

Literature Review

The application of Artificial Intelligence (AI) and Machine Learning (ML) in educational systems has rapidly expanded over the past decade, shifting from basic statistical analysis to highly complex predictive modeling. This chapter extensively reviews the existing literature surrounding educational data mining, the mathematical evolution of multi-label (Tsoumakas & Katakis, 2007) classification algorithms, and the critical, emerging need for Explainable AI (XAI) in high-stakes guidance systems. By critically examining the current state of academic research, this chapter establishes the robust theoretical foundation required for transitioning from simple, isolated academic performance prediction to an intelligent, mathematically explainable, and multi-pathway career recommendation framework.

Theoretical Framework: From Academic Performance to Career Pathway

Traditionally, educational guidance has been viewed by both educators and sociologists as a highly linear, deterministic progression: a student's past academic performance dictates their immediate educational choices, which subsequently permanently restrict or open up future career paths. Educational Data Mining (Romero & Ventura, 2010) (EDM) emerged as a formalized discipline precisely to quantify and analyze these transitions. According to foundational work by Romero and Ventura (2010), EDM focuses on developing mathematical, statistical, and computational methods to discover unique, actionable patterns in the massive amounts of data coming from educational environments.

The theoretical framework underpinning this thesis rejects the linear, single-destination model and instead proposes a continuous, intelligent, and multi-branching pathway:

$$\text{Academic Performance (Input)} \xrightarrow{\text{ML Prediction}} \text{Educational Pathway} \xrightarrow{\text{Recommendation}} \text{Career Pathway}$$

In this mathematical context, the input consists of numerical scores from standardized tests.

Let X represent the high-dimensional feature space. For the scope of this thesis, $X = [x_{math}, x_{phys}, x_{bio}, x_{chem}]^T$, representing the core STEM subjects required for health sciences. The traditional, outdated algorithmic approach is to map X to a single target variable Y (a specific, isolated career).

However, human potential, cognitive aptitude, and professional capabilities are inherently not mutually exclusive. A student excelling simultaneously in Biology and Chemistry could successfully pursue Medicine, Pharmacy, or Nursing with equal theoretical probability. Therefore, the critical theoretical shift required in modern EDM is moving away from the restrictive mapping $f : X \to Y$ and toward a multi-dimensional mapping $f : X \to \{0, 1\}^K$, where K represents the total number of possible, overlapping career paths (Baker & Inventado, 2014). This framework forms the absolute core of the thesis.

Chronological Evolution of Educational Guidance Systems

The methodology for guiding students toward appropriate career pathways has undergone a radical transformation over the past century, evolving from entirely subjective human assessments to highly mathematical, data-driven automated systems.

In the late 20th century, guidance was fundamentally reliant on psychometric testing and manual counselor evaluation. Tools such as the Holland Occupational Themes (RIASEC) provided a structured, albeit entirely manual, framework for categorizing student personalities and matching them to predefined career clusters. While foundational, these systems were static. They relied on a student's self-reported psychological state at a single point in time, completely ignoring the longitudinal, empirical reality of their hard academic performance in subjects like Mathematics and Physics.

By the early 2000s, the advent of basic database management systems allowed educational institutions to deploy "Expert Systems." These early computational models utilized hard-coded, rules-based logic (e.g., IF Math > 80 AND Biology > 75 THEN Recommend Pre-Med). However, expert systems suffered from a fatal flaw known as the "knowledge acquisition bottleneck." Because human experts had to manually define every single rule and threshold, the systems were incredibly brittle, entirely unable to scale, and failed to capture the non-linear, complex covariance between different academic disciplines that define true cognitive aptitude.

It was not until the mass digitization of national educational records (Educational Data Mining) in the 2010s that true Machine Learning could be applied to this domain, shifting the paradigm from manually written rules to autonomously discovered mathematical patterns.

Machine Learning in Educational Data Mining

Predictive modeling in education has a rich history of utilizing historical data to forecast student success, retention, or dropout rates. Early academic studies primarily relied on foundational statistical methods such as logistic regression and basic, singular decision trees. While these models are highly interpretable, meaning human operators can easily extract the exact mathematical coefficients or rules dictating how the model makes decisions, they fundamentally fail to capture the complex, highly non-linear interactions between different academic subjects that truly define student capability.

For example, a standard Decision Tree partitions the feature space recursively using orthogonal hyperplanes. If x_{math} represents the Mathematics score and x_{bio} represents the Biology score, the tree uses a simple splitting criterion (such as Gini Impurity) to create a rule such as:

$$\text{If } x_{math} > 70 \text{ and } x_{bio} > 75 \implies \text{Recommend Medicine}$$

While this is mathematically simple and logically easy to understand for a human counselor, this rigid, box-like structure often leads to "overfitting" in complex educational datasets. The model learns the exact statistical noise of the training data rather than the underlying true pedagogical pattern. To overcome this critical limitation, EDM researchers began heavily utilizing ensemble methods. Ensemble learning leverages the mathematical principle of the "wisdom of crowds," combining multiple weak, high-variance models to create a single strong, highly generalized predictive model (Romero & Ventura, 2010).

Single-Label vs. Multi-Label (Tsoumakas & Katakis, 2007) Classification

To fully appreciate the methodology of this thesis, one must understand the stark mathematical differences between traditional single-label classification and multi-label (Tsoumakas & Katakis, 2007) classification.

In standard single-label classification, the objective is to learn a hypothesis function that assigns a given instance x_i to exactly one class out of a set of mutually exclusive classes. Mathematically, if $L = \{l_1, l_2, \ldots, l_K\}$ is a finite, predefined set of labels, a single-label model predicts a single output $y_i \in L$. The probabilities across all classes must sum exactly to 1:

$$\sum_{k=1}^{K} P(y = l_k | x) = 1$$

However, as established in the theoretical framework, health sector career guidance is inherently a **Multi-Label Classification** problem. A student possesses a complex skill profile that makes them suitable for multiple careers simultaneously. In multi-label (Tsoumakas & Katakis, 2007) classification, the model is tasked with predicting a subset of labels $Y \subseteq L$ for each instance. This means the output is no longer a single scalar, but rather a vector $Y = [y_1, y_2, \ldots, y_K]$, where $y_k \in \{0, 1\}$ indicates the relevance or recommendation of the k-th career (Tsoumakas & Katakis, 2007). In this paradigm, the probabilities for each individual label are calculated independently, and their sum is not constrained to 1.

Zhang and Zhou (2013) comprehensively categorize multi-label (Tsoumakas & Katakis, 2007) learning algorithms into two primary, distinct approaches:

1. **Problem Transformation Methods:** These methods mathematically convert the complex multi-label (Tsoumakas & Katakis, 2007) problem into one or more traditional single-label problems, allowing standard algorithms (like basic SVMs or Decision Trees) to be used without internal modification.

2. **Algorithm Adaptation Methods:** These methods rewrite the core internal mathematics of existing algorithms (such as altering the entropy calculation of a Decision Tree) to allow them to directly handle multi-label (Tsoumakas & Katakis, 2007) data.

This thesis focuses heavily on Problem Transformation Methods, specifically seeking to overcome the limitations of the most basic transformation: Binary Relevance.

Binary Relevance and the Independence Assumption

The simplest, most naive problem transformation method is **Binary Relevance (BR)**. The BR algorithm operates by training K completely independent binary classifiers, one for each distinct label in the dataset. For a new instance x, the final multi-label (Tsoumakas & Katakis, 2007) prediction is simply the logical union of the positive labels predicted by all K individual classifiers.

While computationally fast and mathematically simple to implement, BR suffers from a fatal theoretical flaw: it assumes that all labels are completely statistically independent. Mathematically, it assumes:

$$P(y_1, y_2, \ldots, y_K|x) = \prod_{k=1}^{K} P(y_k|x)$$

In the specific context of guiding students into highly specialized health sector careers, this independence assumption spectacularly fails. The probability of a student being suited for "Nursing" is highly positively correlated with the probability of them being suited for "Midwifery." If an intelligent model predicts a high likelihood for one, it should mathematically leverage that newly discovered information to increase the predicted probability of the other. BR ignores this entirely.

Classifier Chains for Label Correlation

To theoretically and mathematically overcome the naive independence assumption of Binary Relevance, Read et al. (2011) introduced the highly effective **Classifier Chains (CC)** method. CC elegantly transforms the multi-label (Tsoumakas & Katakis, 2007) problem into a sequential chain of binary classifiers. The critical innovation is that subsequent classifiers in the chain utilize the actual predictions of preceding classifiers as additional input features, thereby forcing the model to learn the correlations between labels.

Let the original input feature vector be x. For K target labels, a sequence of K classifiers $C_1, C_2, \ldots, C_K$ is trained. The training process proceeds as follows:

- Classifier C_1 is trained on the original features x to predict label y_1.
- Classifier C_2 is trained on an extended feature space consisting of x appended with the true value of y_1 (i.e., $[x, y_1]$) to predict y_2.
- Generalizing this, Classifier C_k is trained on the feature space $[x, y_1, y_2, \ldots, y_{k-1}]$ to predict y_k.

During the inference (prediction) phase on a new, unseen student, the true labels y are completely unknown. Therefore, the predictions $\hat{y}$ from previous classifiers in the chain are passed down instead:

$$\hat{y}_k = C_k(x, \hat{y}_1, \ldots, \hat{y}_{k-1})$$

This chaining mechanism fundamentally changes the probability calculation. Instead of independent probabilities, CC models the joint probability of the labelset using the chain rule of probability:

$$P(y_1, y_2, \ldots, y_K|x) = P(y_1|x) \cdot P(y_2|x, y_1) \cdot \cdots \cdot P(y_K|x, y_1, \ldots, y_{k-1})$$

This allows the model to capture deep, complex correlations between different career paths, making it vastly more suitable for intelligent educational guidance than standard Binary Relevance.

Review of Selected Base Algorithms

Within the Classifier Chain (Read et al., 2011) framework, a "base algorithm" must be chosen to act as the individual binary classifier (C_k) at each link in the chain. The mathematical complexity and learning capability of the CC model depend entirely on this base algorithm. This thesis explores four powerful, mathematically distinct algorithms:

Decision Trees (DT)

A Decision Tree recursively splits the input data space based on the feature that maximizes a specific purity metric, typically Information Gain. If a dataset D has C classes, the Shannon entropy $H(D)$ is defined mathematically as:

$$H(D) = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

where p_i is the empirical proportion of samples belonging to class i. The Information Gain for a split on feature A is the difference between the parent node's entropy and the weighted sum of the child nodes' entropies. While highly interpretable, singular decision trees are incredibly prone to high variance and severe overfitting when dealing with deep, continuous numerical data like exam scores.

Random Forest (RF)

To solve the high variance problem of single decision trees, Breiman (2001) introduced Random Forests. A Random Forest (Breiman, 2001) is a sophisticated ensemble algorithm that combines the concepts of "bagging" (bootstrap aggregating) and random feature selection. It builds hundreds of deep, independent decision trees in parallel. Each tree is trained on a random bootstrap sample of the data, and at each split in the tree, only a random subset of features is considered. The final prediction for a regression or probability task is made by averaging the outputs of all trees:

$$\hat{f}_{RF}(x) = \frac{1}{B}\sum_{b=1}^{B} f_b(x)$$

where B is the total number of trees. This mathematical averaging heavily reduces the variance of the model without significantly increasing the bias, making RF exceptionally robust against overfitting.

XGBoost (Extreme Gradient Boosting)

XGBoost, developed by Chen and Guestrin (2016), represents the absolute state-of-the-art for structured, tabular data analysis. Unlike Random Forest (Breiman, 2001), which builds deep trees independently in parallel, XGBoost (Chen & Guestrin, 2016) builds shallow trees sequentially. This is the core concept of "boosting." Each new tree is specifically trained to focus on correcting the mathematical errors (the residuals or gradients) made by the combination of all previous trees.

The objective function optimized at iteration t contains a specific loss function l (which measures how far the prediction is from the truth) and a regularization term Ω (which penalizes overly complex trees to prevent overfitting):

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$$

where the regularization term is defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

Here, T is the number of leaves in the tree, w_j are the leaf weights, and γ and λ are hyperparameter constants. By utilizing a second-order Taylor expansion to approximate the loss function, XGBoost (Chen & Guestrin, 2016) optimizes this objective with extreme computational efficiency, making it a highly dominant algorithm in modern data science competitions and enterprise applications.

Multi-Layer Perceptron (MLP)

An MLP is the foundational architecture for deep artificial neural networks. It consists of an input layer, one or more hidden layers, and an output layer. Neurons in one layer are densely connected to neurons in the next, with each connection governed by a mathematical weight w_{ij}. The output z_j of a neuron j is calculated by applying a non-linear activation function σ to the weighted sum of all its inputs plus a bias term b_j:

$$z_j = \sigma\left(\sum_i w_{ij} x_i + b_j\right)$$

For modern networks, the activation function σ is typically the Rectified Linear Unit (ReLU), defined simply as $\max(0, x)$. The network learns by utilizing the backpropagation algorithm to calculate the gradient of the loss function with respect to every weight, updating the weights using optimizers like Stochastic Gradient Descent (SGD) or Adam. While MLPs possess the theoretical capacity (via the Universal Approximation Theorem) to model any continuous non-linear relationship, their deeply layered nature makes them the ultimate "black box," rendering it nearly impossible for a human educator to understand exactly how a recommendation was reached without external XAI tools.

Limitations of Traditional Algorithms (SVM and Naive Bayes)

While the aforementioned tree-based ensembles (XGBoost, RF) and MLPs represent the cutting edge for this specific problem, much of the early literature in educational data mining heavily relied on Support Vector Machines (SVM) and Naive Bayes classifiers. It is mathematically critical to understand why these algorithms are inherently sub-optimal for multi-label (Tsoumakas & Katakis, 2007) health sector guidance based on raw exam scores.

Naive Bayes (NB) operates on the strict mathematical assumption of conditional independence between features given the class label:

$$P(x_{math}, x_{bio}, x_{chem}|Y = 1) = P(x_{math}|Y = 1) \cdot P(x_{bio}|Y = 1) \cdot P(x_{chem}|Y = 1)$$

In reality, a student's performance in Physics is highly correlated with their performance in Mathematics. The Naive Bayes algorithm completely ignores this deep covariance matrix, leading to severely biased probability estimates and poor ranking performance in multi-label (Tsoumakas & Katakis, 2007) scenarios.

Support Vector Machines (SVM), conversely, attempt to find a hyper-plane in an N-dimensional space that distinctly classifies the data points. While mathematically elegant (often utilizing the "kernel trick" to map data into higher dimensions for linear separability), SVMs are computationally expensive and scale very poorly ($O(n^3)$) with large educational datasets. Furthermore, SVMs natively output distance metrics to a hyperplane, not true probabilities. To generate the necessary ranked probabilities for career recommendations, SVM outputs must be artificially squeezed through Platt scaling (a logistic regression model trained on the SVM scores), which introduces an additional layer of unnecessary approximation error compared to algorithms like Random Forest (Breiman, 2001) that natively calculate empirical probabilities based on leaf-node distributions.

Explainable AI (XAI) and Counterfactuals

The rapid deployment of advanced ML systems (like XGBoost (Chen & Guestrin, 2016) and MLPs) in high-stakes socio-technical areas such as healthcare, criminal justice, and education demands absolute algorithmic transparency. A career guidance model that simply outputs a probability matrix indicating "Recommend Medicine: 92%" without any justification is highly unlikely to be trusted or adopted by students, parents, or government ministries.

Explainable AI (XAI) seeks to penetrate the black box (Wachter et al., 2017), providing mathematical visibility into the decision-making process of complex ensemble and deep learning models. Foundational work by Lundberg and Lee (2017) proposed SHAP (SHapley Additive exPlanations), a rigorous game-theoretic approach that assigns a specific, mathematically guaranteed contribution value to each feature (e.g., Mathematics score vs. Biology score) for every specific prediction made by the model.

However, knowing *why* a decision was made is often insufficient for human users seeking guidance. Furthermore, Wachter et al. (2017) introduced the powerful concept of **Counterfactual Explanations**. Instead of just explaining why a model rejected a student for a specific career, counterfactuals mathematically explain exactly what would need to change in the student's profile to receive the desired decision.

As further discussed by Barocas et al. (2020), counterfactuals are highly actionable. Mathematically, finding a counterfactual (Wachter et al., 2017) x^* for an original, factual input x involves setting up an optimization problem to minimize the distance d between them, subject to the strict constraint that the model's prediction output changes to the desired class:

$$x^* = \arg\min_{x'} d(x, x') \quad \text{subject to } f(x') \neq f(x)$$

In a practical educational context, this advanced mathematical formulation allows the guidance system to generate statements such as: *"You were recommended for Nursing based on your current profile. However, if your Chemistry raw mark had been exactly 12 points higher, the algorithm would have shifted your primary recommendation to Pharmacy."* This level of actionable, personalized insight is entirely absent from current educational data mining applications.

Ethical AI and Algorithmic Fairness in Education

As AI systems move from academic research to physical deployment in national education infrastructure, the mathematical concept of "fairness" becomes a critical limitation that must be addressed. A model trained on historical data is inherently at risk of learning and perpetuating historical biases (Barocas & Selbst, 2016).

If, historically, female students were disproportionately discouraged from pursuing advanced degrees in STEM, a naive machine learning algorithm will observe that correlation in the training data and incorrectly learn that gender is a negative predictive feature for success in medicine. This results in algorithmic discrimination.

To counteract this, modern ethical AI frameworks require that models achieve *Demographic Parity* or *Equalized Odds*. Mathematically, Demographic Parity requires that the probability of a positive recommendation ($\hat{Y} = 1$) is completely statistically independent of a protected attribute A (such as gender or socio-economic background):

$$P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$$

While this thesis specifically mitigates gender and socio-economic bias by completely omitting those variables from the input feature space (relying strictly on objective, anonymized examination scores $x_{math}, x_{phys}, x_{bio}, x_{chem}$), acknowledging the mathematical constraints of algorithmic fairness remains a critical component of deploying ethical educational AI.

Research Gap

While considerable academic research currently exists on predicting binary student outcomes (such as passing or failing, dropout rates) and applying basic algorithms like standard Decision Trees to single-label career choices, a comprehensive, multi-label (Tsoumakas & Katakis, 2007) approach seamlessly integrated with deep mathematical explainability remains incredibly scarce, particularly within the context of the Rwandan educational system.

Previous systems developed in this domain act as isolated, predictive black boxes, entirely neglecting the crucial, human-centric steps of nuanced recommendation and actionable explanation. By rigorously integrating the Classifier Chains (Read et al., 2011) problem transformation methodology, advanced state-of-the-art ensemble algorithms (XGBoost), and the foundational mathematics of counterfactual (Wachter et al., 2017) logic, this thesis seeks to bridge that immense theoretical and practical gap. The result is the proposition of a highly transparent, mathematically rigorous, and multi-pathway guidance model designed for real-world educational deployment.

Deep Mathematical Foundations of Tree-Based Ensembles

To fully contextualize the methodological superiority of the eXtreme Gradient Boosting (XGBoost) algorithm employed in this research, an exhaustive review of the underlying mathematical foundations of decision tree ensembles is required.

Information Theory and Shannon Entropy

The fundamental building block of any decision tree is the recursive partitioning of the feature space. This partitioning is guided by Information Theory, specifically the concept of Shannon Entropy, introduced by Claude Shannon in 1948. Entropy $H(S)$ measures the impurity or uncertainty of a dataset S with respect to the target variable y. For a multi-label (Tsoumakas &

Katakis, 2007) context with classes C, the entropy is defined as:

$$H(S) = -\sum_{i=1}^{|C|} p_i \log_2(p_i) \tag{1}$$

where p_i is the empirical probability of a sample belonging to class i within the subset S. A node is considered "pure" if $H(S) = 0$, meaning all samples belong to the exact same class configuration. The objective of the decision tree splitting criterion is to maximize the Information Gain (IG), which is the reduction in entropy achieved by partitioning S according to a specific feature threshold A:

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v) \tag{2}$$

Gini Impurity vs. Entropy

While Shannon Entropy provides a robust theoretical foundation, many modern implementations, such as Random Forest (Breiman, 2001) (Breiman, 2001), utilize Gini Impurity due to its computational efficiency. Gini Impurity $G(S)$ measures the probability that a randomly chosen element from the set would be incorrectly labeled if it was randomly labeled according to the distribution of labels in the subset:

$$G(S) = 1 - \sum_{i=1}^{|C|} p_i^2 \tag{3}$$

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Unlike Random Forests which rely on Bagging (Bootstrap Aggregating) to reduce variance by averaging parallel trees, XGBoost (Chen & Guestrin, 2016) utilizes Boosting to reduce bias by sequentially training trees to correct the residual errors of preceding trees. The objective function

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XGBoost optimizes this objective by taking a second-order Taylor expansion around the current estimate, functionally executing a Newton-Raphson step in function space:

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^{n} \left[l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \tag{5}$$

where g_i and h_i are the first and second order gradient statistics on the loss function, respectively. This rigorous mathematical foundation is exactly why XGBoost (Chen & Guestrin, 2016) outperforms the Multi-Layer Perceptron (Zhang & Zhou, 2013) on the sharp, threshold-based synthetic data generated for this thesis.

Explainable AI (XAI): From LIME (Lundberg & Lee, 2017) to Counterfactuals

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Counterfactual Optimization Bounds

Counterfactuals solve this by posing the question: "What is the minimum alteration required to flip the prediction?" Mathematically, this is an optimization problem seeking the perturbation vector δ:

$$\delta^* = \arg\min_{\delta} ||\delta||_p \quad \text{subject to} \quad f(x+\delta) = y_{target} \text{ and } x+\delta \in \mathcal{F} \tag{6}$$

where $||\cdot||_p$ is a distance metric (commonly L_1 or L_2) and $\mathcal{F}$ represents the space of feasible actions (e.g., preventing negative changes, as a student cannot deliberately lower their past scores to achieve a better outcome).

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While Shannon Entropy provides a robust theoretical foundation, many modern implementations, such as Random Forest (Breiman, 2001) (Breiman, 2001), utilize Gini Impurity due to its computational efficiency. Gini Impurity $G(S)$ measures the probability that a randomly chosen element from the set would be incorrectly labeled if it was randomly labeled according to the distribution of labels in the subset:

$$G(S) = 1 - \sum_{i=1}^{|C|} p_i^2 \tag{75}$$

The gradient of the Gini index is computationally cheaper to calculate because it avoids the logarithmic operations required by Shannon Entropy.

Gradient Boosting Optimization and Newton-Raphson Step

Unlike Random Forests which rely on Bagging (Bootstrap Aggregating) to reduce variance by averaging parallel trees, XGBoost (Chen & Guestrin, 2016) utilizes Boosting to reduce bias by sequentially training trees to correct the residual errors of preceding trees. The objective function

$\mathcal{L}^{(t)}$ at iteration t contains both a loss term l and a regularization term Ω:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \tag{76}$$

XGBoost optimizes this objective by taking a second-order Taylor expansion around the current estimate, functionally executing a Newton-Raphson step in function space:

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^{n} \left[l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \tag{77}$$

where g_i and h_i are the first and second order gradient statistics on the loss function, respectively. This rigorous mathematical foundation is exactly why XGBoost (Chen & Guestrin, 2016) outperforms the Multi-Layer Perceptron (Zhang & Zhou, 2013) on the sharp, threshold-based synthetic data generated for this thesis.

Explainable AI (XAI): From LIME (Lundberg & Lee, 2017) to Counterfactuals

The deployment of machine learning in critical public sectors, such as education and career guidance, necessitates transparency. The European Union's General Data Protection Regulation (GDPR) explicitly mandates a "right to explanation" for algorithmic decisions.

Local Interpretable Model-agnostic Explanations (LIME)

LIME operates by perturbing the input data and fitting a linear surrogate model around the local vicinity of the prediction. While LIME (Lundberg & Lee, 2017) provides feature attributions, it fails to provide actionable pathways. A student informed that "Biology contributed 40% to your rejection" gains no pedagogical recourse.

Counterfactual Optimization Bounds

Counterfactuals solve this by posing the question: "What is the minimum alteration required to flip the prediction?" Mathematically, this is an optimization problem seeking the perturbation vector δ:

$$\delta^* = \arg\min_{\delta} ||\delta||_p \quad \text{subject to} \quad f(x+\delta) = y_{target} \text{ and } x+\delta \in \mathcal{F} \tag{78}$$

where $||\cdot||_p$ is a distance metric (commonly L_1 or L_2) and $\mathcal{F}$ represents the space of feasible actions (e.g., preventing negative changes, as a student cannot deliberately lower their past scores to achieve a better outcome).

Deep Mathematical Foundations of Tree-Based Ensembles

To fully contextualize the methodological superiority of the eXtreme Gradient Boosting (XGBoost) algorithm employed in this research, an exhaustive review of the underlying mathematical foundations of decision tree ensembles is required.

Information Theory and Shannon Entropy

The fundamental building block of any decision tree is the recursive partitioning of the feature space. This partitioning is guided by Information Theory, specifically the concept of Shannon Entropy, introduced by Claude Shannon in 1948. Entropy $H(S)$ measures the impurity or uncertainty of a dataset S with respect to the target variable y. For a multi-label (Tsoumakas & Katakis, 2007) context with classes C, the entropy is defined as:

$$H(S) = -\sum_{i=1}^{|C|} p_i \log_2(p_i) \tag{79}$$

where p_i is the empirical probability of a sample belonging to class i within the subset S. A node is considered "pure" if $H(S) = 0$, meaning all samples belong to the exact same class configuration. The objective of the decision tree splitting criterion is to maximize the Information

Gain (IG), which is the reduction in entropy achieved by partitioning S according to a specific feature threshold A:

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where $||\cdot||_p$ is a distance metric (commonly L_1 or L_2) and $\mathcal{F}$ represents the space of feasible actions (e.g., preventing negative changes, as a student cannot deliberately lower their past scores to achieve a better outcome).

Methodology

This chapter exhaustively details the mathematical, computational, and operational methodological approach employed to construct the explainable multi-label (Tsoumakas & Katakis, 2007) career guidance model. It rigorously outlines the quantitative research design, the specifics of data collection and preprocessing strategies, the deep architectural configuration of the machine learning algorithms, the mathematical formulation of the explainability layer, and the multi-label (Tsoumakas & Katakis, 2007) evaluation metrics used to assess the model's performance.

Research Design

This study adopts a highly structured quantitative, predictive modeling research design grounded in the principles of computational data science. It relies heavily on secondary, historical, and numerical continuous data collected directly from national examination records to train advanced machine learning models.

The research follows a strict, systematic data science lifecycle architecture:

1. **Data Acquisition & Aggregation:** Compiling disjointed yearly datasets into a singular longitudinal database.
2. **Exploratory Data Analysis (EDA):** Statistically analyzing feature distributions, correlations, and identifying systemic outliers.
3. **Data Preprocessing & Feature Engineering:** Handling missing values, normalizing feature vectors, and transforming text targets into mathematical multi-hot encoded vectors.
4. **Model Training & Hyperparameter Tuning:** Implementing the Classifier Chains (Read et al., 2011) meta-algorithm and tuning the base learners (XGBoost, RF, DT, MLP).

5. **Explainability Integration (XAI):** Extracting global feature importance and generating local counterfactual (Wachter et al., 2017) explanations.

6. **Mathematical Evaluation:** Assessing the algorithms against strict multi-label (Tsoumakas & Katakis, 2007) mathematical metrics rather than simple accuracy.

The design specifically transitions from a traditional single-target, mutually exclusive prediction framework to a comprehensive multi-target (multi-label) probabilistic ranking framework.

Data Collection and Description

The primary dataset utilized in this research consists of Rwandan Ordinary Level (S3) national examination records, generously aggregated over an extended longitudinal period spanning from 2017 to 2025. This multi-year aggregation provides a highly robust volume of data, comprising thousands of individual student records, which is absolutely necessary for training deep, advanced machine learning algorithms without severe overfitting.

The dataset primarily focuses on the raw numerical marks obtained by students in the core STEM subjects deemed strictly essential for advanced health sector educational pathways. The key input feature vector (X), mathematically represented as $x_i \in \mathbb{R}^4$, extracted from the dataset includes:

- x_1: **Biology and Health Sciences raw marks**
- x_2: **Chemistry raw marks**
- x_3: **Mathematics raw marks**
- x_4: **Physics raw marks**

Additional administrative and demographic features such as *candidate_code*, *School code*, and *level* are retained in the raw database for unique identification and potential future bias analysis. However, they are strictly excluded from the mathematical model training matrix to actively prevent the algorithm from learning geographic, institutional, or socioeconomic biases, ensuring the predictions are based purely on academic cognitive merit.

Data Preprocessing and Target Transformation

Raw educational data exported from government databases is rarely, if ever, structurally suitable for immediate algorithmic training. The data preprocessing pipeline involves several critical mathematical and programmatic steps to ensure absolute data integrity and structural model compatibility.

Handling Missing Values and Normalization

Any student records containing missing values in the core subject continuous columns are algorithmically imputed. Rather than using the dataset mean, which is highly sensitive to extreme outliers, missing values are imputed using the median score of that specific year to preserve the true statistical distribution. Records that are heavily corrupted or missing more than 50% of their features are dropped entirely from the matrix.

Because specific gradient-based optimization algorithms (like the Stochastic Gradient Descent used in the Multi-Layer Perceptron) are highly sensitive to the magnitude and scale of input features, the raw marks are mathematically normalized. Min-Max scaling is applied to linearly transform all subject scores to a uniform scale bound strictly between 0 and 1:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

While tree-based ensemble models (XGBoost, Random Forest) are scale-invariant and do not strictly require this normalization to split nodes, maintaining a uniform scale across all models is mathematically essential for generating consistent distance calculations ($L2$ norms) during the counterfactual (Wachter et al., 2017) explanation generation later in the pipeline.

Covariance and Feature Collinearity Analysis

Before feeding the normalized matrix into the multi-label (Tsoumakas & Katakis, 2007) models, the internal relationships between the predictive features are rigorously analyzed via a Covariance Matrix (Σ) and Pearson Correlation Coefficients (r). For any two features X and Y (e.g., Mathematics and Physics), the sample covariance is computed as:

$$Cov(X,Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

And the normalized Pearson correlation coefficient is:

$$r_{xy} = \frac{Cov(X,Y)}{s_x s_y}$$

Where s_x and s_y are the sample standard deviations. Identifying high positive collinearity (e.g., $r > 0.8$) between specific subjects computationally validates the decision to use Classifier Chains (Read et al., 2011), as it empirically proves that a student's cognitive capabilities are not independent silos, but rather highly correlated, intersecting skill sets.

Multi-Label Target Encoding (Multi-Hot Encoding)

In a traditional dataset structure, a student might have a single string label indicating their final chosen career (e.g., "Nurse"). To train a multi-label (Tsoumakas & Katakis, 2007) algorithmic model, these string targets must be mathematically transformed into vectors. Let K be the total predefined number of possible, distinct health sector careers (e.g., $l_1 =$ Medicine, $l_2 =$ Nursing, $l_3 =$ Pharmacy, $l_4 =$ Midwifery, $l_5 =$ Public Health).

The target output for each student is mathematically encoded as a binary vector $Y \in \{0,1\}^K$. For instance, if a student's rigorous academic profile mathematically qualifies them for both Medicine and Pharmacy, but not the others, their target vector would be encoded as:

$$Y = [1, 0, 1, 0, 0]$$

This multi-hot encoding allows the loss functions within the algorithms to mathematically penalize and learn the overlapping, non-exclusive academic requirements for different careers simultaneously.

Model Architecture and Implementation

The algorithmic core of this methodology is the application of the **Classifier Chains (CC)** meta-algorithm combined with highly optimized base learners.

The Classifier Chain (Read et al., 2011) Mechanism

As rigorously discussed in the literature review, career compatibilities are highly statistically correlated. The CC technique accounts for this dependency by training K sequential binary classifiers. The order of the chain is a critical hyperparameter that can be statically defined, optimized via grid search, or randomly ensembled (creating an Ensemble of Classifier Chains (Read et al., 2011) or ECC).

For a given student input feature vector x, the sequential probabilistic calculation proceeds as follows:

1. The first classifier calculates the probability of the first career given the features: $P(y_1|x)$.
2. The second classifier calculates the probability of the second career given both the features and the prediction of the first: $P(y_2|x, y_1)$.
3. The final classifier calculates the conditional probability given all previous predictions: $P(y_K|x, y_1, y_2, \ldots, y_{K-1})$.

This sequential mechanism mathematically ensures that if a student is highly recommended for one highly demanding medical field (e.g., Medicine), the probabilities for heavily related fields (e.g., Pharmacy) are mathematically adjusted upwards based on the learned correlations within the chain.

Base Algorithms Configuration

Four mathematically distinct algorithms are utilized, tuned, and compared as the base classifiers (C_k) within the chain. Their internal mathematical mechanisms dictate their performance and ability to model the non-linear feature space:

1. **Decision Tree (DT):** Serves as a baseline, highly interpretable model. The algorithm recursively partitions the feature space. At each node m, representing a region R_m with N_m observations, the algorithm searches for a splitting variable j and split point s that minimizes the Gini impurity. If p_{mk} is the proportion of class k observations in node m, the Gini index is defined as:

$$G(m) = \sum_{k=1}^{K} p_{mk}(1 - p_{mk})$$

   The split (j, s) is chosen to maximize the impurity decrease: $\Delta G = G(parent) - (w_{left}G(left) + w_{right}G(right))$. Hyperparameters such as `max_depth` and `min_samples_split` are strictly constrained to prevent catastrophic overfitting on continuous data.

2. **Random Forest (RF):** An advanced ensemble method that massively reduces the variance of individual decision trees by applying bootstrap aggregating (bagging) and random feature subspace selection. For B trees, a bootstrap sample Z^{*b} of size N is drawn from the training data. A tree T_b is grown, but at each split, only a random subset of $m \approx \sqrt{p}$ features is considered. The final probability prediction for a class k given input x is the

averaged vote:

$$\hat{P}_{RF}(Y = k|x) = \frac{1}{B}\sum_{b=1}^{B}\hat{P}_{T_b}(Y = k|x)$$

By injecting randomness, the trees become mathematically de-correlated, forcing diversity and creating an incredibly robust model against statistical noise.

3. **XGBoost (eXtreme Gradient Boosting):** A highly optimized, regularized boosting algorithm that builds trees sequentially to minimize the residual errors of prior trees. At step t, it seeks to add a tree f_t that minimizes the objective:

$$\text{Obj}^{(t)} = \sum_{i=1}^{n} L\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$$

XGBoost (Chen & Guestrin, 2016) utilizes a second-order Taylor expansion to approximate the loss function L, computing the first derivative (gradient) $g_i = \partial_{\hat{y}_i^{(t-1)}} L(y_i, \hat{y}_i^{(t-1)})$ and second derivative (hessian) $h_i = \partial^2_{\hat{y}_i^{(t-1)}} L(y_i, \hat{y}_i^{(t-1)})$. The optimal leaf weight w_j^* for leaf j containing instance set I_j is mathematically derived as:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}$$

where λ is the $L2$ regularization term preventing overly large weights. The learning rate (η) scales this weight addition to prevent early overfitting.

4. **Multi-Layer Perceptron (MLP):** A deep feedforward neural network architecture. The network is configured with multiple hidden layers utilizing the Rectified Linear Unit (ReLU) activation function: $a^{(l)} = \max(0, W^{(l)}a^{(l-1)} + b^{(l)})$. The output layer utilizes the Sigmoid activation function $\sigma(z) = \frac{1}{1+e^{-z}}$ to map raw logits to probability bounds $[0, 1]$. The network minimizes the Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

The Adam optimizer computes gradients via the chain rule (backpropagation). For a weight $w_{jk}^{(l)}$, the gradient update utilizes first and second moment estimates of the gradient:

$$w_{jk}^{(l)} \leftarrow w_{jk}^{(l)} - \alpha \frac{\hat{m}}{\sqrt{\hat{v}} + \epsilon}$$

where $\hat{m}$ and $\hat{v}$ are the bias-corrected moving averages of the gradient and squared gradient, respectively. This allows for highly adaptive, automated learning rates on a per-parameter basis.

The Intelligence Layers: Explainability and Counterfactuals

A paramount objective of this thesis is to shatter the black-box nature of advanced ML models. Two specific XAI techniques are mathematically implemented:

Global Explanation: Feature Importance Ranking

For the tree-based ensemble models (Random Forest and XGBoost), global feature importance is mathematically calculated by tracking how much each specific feature (e.g., the Mathematics score) decreases the overall impurity (calculated via Gini impurity or entropy) across all nodes in all trees where that feature is used to split the data. The total decrease in node impurity is weighted by the probability of reaching that node. This allows the system to generate a highly accurate global explanation, empirically proving which STEM subjects dictate success in specific health careers.

Local Explanation: Counterfactual (Wachter et al., 2017) Generation

For local explainability (explaining a single student's unique recommendation), the system utilizes a complex mathematical search to generate a counterfactual (Wachter et al., 2017). If a student strongly desires to study Medicine but is rejected by the model ($f(x) = 0$), the algorithm computationally searches for the closest theoretical, fictional student profile x^* that *is* accepted

($f(x^*) = 1$).

This is formulated strictly as a constrained mathematical optimization problem. The primary objective is to minimize the distance $d(x, x')$ between the student's actual normalized scores x and the required theoretical scores x'. While the Manhattan distance ($L1$ norm, defined as $\sum |x_i - x'_i|$) promotes sparsity (changing only one subject at a time), this thesis specifically utilizes the Euclidean distance ($L2$ norm) to encourage the model to suggest smaller, simultaneous improvements across multiple subjects, which is far more realistic pedagogically than expecting a massive jump in a single subject.

The optimization equation is formulated as:

$$x^* = \arg\min_{x'} \left(\sqrt{\sum_{i=1}^{m} (x_i - x'_i)^2} \right) + \lambda \cdot \text{Cost}(x, x')$$

Subject to the absolute constraint: $f(x') = 1$ (The model prediction must flip to positive).

The additional penalty term, $\lambda \cdot \text{Cost}(x, x')$, is critical for ensuring the generated counterfactual (Wachter et al., 2017) is practically actionable in the real world. For example, it is mathematically impossible for a student to travel back in time and lower their score. Therefore, the cost function heavily penalizes or entirely forbids negative perturbations (where $x'_i < x_i$).

By computationally solving this constrained optimization space using gradient-free search algorithms (as tree-based models are non-differentiable), the system outputs precise, mathematically verified advice: *"To achieve a recommendation for Medicine, you must improve your normalized Chemistry score by 0.15 (15%) and your Biology score by 0.05 (5%)."*

Evaluation Metrics

Evaluating a multi-label (Tsoumakas & Katakis, 2007) classification model requires highly specific, multi-dimensional mathematical metrics, as standard single-label accuracy is fundamentally insufficient and misleading. The algorithms are rigorously evaluated and compared using the following matrix calculations:

- **Hamming Loss:** Measures the fraction of the wrong labels to the total number of labels across the entire dataset. It is highly forgiving of partial correctness. A lower Hamming Loss indicates a superior model.

$$\text{Hamming Loss} = \frac{1}{NL}\sum_{i=1}^{N}\sum_{j=1}^{L}\mathbb{I}(y_{i,j} \neq \hat{y}_{i,j})$$

  where N is the total number of students, L is the number of career labels, and $\mathbb{I}$ is the indicator function that returns 1 if the condition is true and 0 otherwise.

- **Subset Accuracy (Exact Match Ratio):** The absolute strictest metric available. It requires the predicted label vector to be an exact, perfect 1-to-1 match with the true label vector. If a model predicts $[1, 1, 0, 0, 0]$ and the truth is $[1, 0, 0, 0, 0]$, the subset accuracy for that student is 0.

$$\text{Subset Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(Y_i = \hat{Y}_i)$$

- **Micro-Averaged F1-Score:** This metric globally aggregates the true positives, false positives, and false negatives across all classes before computing the harmonic mean of precision and recall. It mathematically ensures that less frequent, highly specialized career paths still factor appropriately into the final evaluation score, heavily penalizing models that only predict the majority class.

$$\text{Micro-F1} = 2 \cdot \frac{\text{Micro-Precision} \cdot \text{Micro-Recall}}{\text{Micro-Precision} + \text{Micro-Recall}}$$

Through the execution of this rigorous, mathematically grounded methodology, the study aims to produce a multi-label (Tsoumakas & Katakis, 2007) model that is highly accurate in prediction, strictly correlated across related career labels via Classifier Chains (Read et al., 2011), and completely transparent in its reasoning via integrated counterfactual (Wachter et al., 2017) optimization.

Algorithmic Pseudocode and Methodological Workflows

To ensure complete reproducibility of the methodological framework, the exact algorithmic pseudocodes utilized in the Python implementations are detailed below.

Classifier Chain Training Protocol

The Classifier Chain (Read et al., 2011) (CC) architecture fundamentally transforms the Multi-Label (Tsoumakas & Katakis, 2007) Classification problem into a sequence of binary classification problems, preserving label dependencies. Let L be the set of labels.

- **Input**: Training data $D = \{(x_i, y_i)\}_{i=1}^{N}$ where $y_i \in \{0, 1\}^{|L|}$.
- **Output**: A sequence of trained binary classifiers $h_1, h_2, \ldots, h_{|L|}$.

The exact mathematical procedure executed by the `02_modeling.py` script is defined as:

1. Define a label ordering. For this thesis, the natural order [Medicine, Pharmacy, Nursing, Public Health, Biomedical] was utilized.
2. For each label j from 1 to $|L|$:
   (a) Create an augmented feature space X'_j by concatenating the original features X with the true labels $Y_{1:j-1}$.
   (b) Train the base estimator h_j (e.g., XGBoost) on the augmented dataset (X'_j, Y_j).
3. During inference, since the true labels are unknown, the prediction $\hat{y}_j$ is iteratively appended to the feature space to predict $\hat{y}_{j+1}$.

Vectorized Counterfactual (Wachter et al., 2017) Search Space

The L_2 minimization algorithm deployed in `03_evaluation_xai.py` utilizes a grid-based constrained search. The complexity of this search is bounded by $\mathcal{O}(S^F)$ where S is the number of discrete step increments and F is the number of mutable features. To prevent combinatorial explosion, the search space was constrained to positive increments (0$\%$ to +50$\%$) in discrete 5$\%$ steps, reducing the search space to a highly efficient, deterministic boundary that guarantees algorithmic convergence within milliseconds.

Architectural Flowchart of Data Processing

The end-to-end data pipeline is structured as a Directed Acyclic Graph (DAG).

1. **Ingestion**: Raw CSV loading via Pandas.
2. **Imputation**: Mean imputation for missing academic scores.
3. **Normalization**: Min-Max scaling to map all features to the L_2 bounded domain $[0, 1]$.
4. **Synthesis**: Boolean logic thresholding to generate the Multi-Label (Tsoumakas & Katakis, 2007) Target Matrix.
5. **Evaluation**: 80/20 train-test stratification to validate out-of-sample generalization.

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The Classifier Chain (Read et al., 2011) (CC) architecture fundamentally transforms the Multi-Label (Tsoumakas & Katakis, 2007) Classification problem into a sequence of binary classification problems, preserving label dependencies. Let L be the set of labels.

- **Input**: Training data $D = \{(x_i, y_i)\}_{i=1}^{N}$ where $y_i \in \{0, 1\}^{|L|}$.
- **Output**: A sequence of trained binary classifiers $h_1, h_2, \ldots, h_{|L|}$.

The exact mathematical procedure executed by the `02_modeling.py` script is defined as:

1. Define a label ordering. For this thesis, the natural order [Medicine, Pharmacy, Nursing, Public Health, Biomedical] was utilized.
2. For each label j from 1 to $|L|$:
   (a) Create an augmented feature space X'_j by concatenating the original features X with the true labels $Y_{1:j-1}$.
   (b) Train the base estimator h_j (e.g., XGBoost) on the augmented dataset (X'_j, Y_j).
3. During inference, since the true labels are unknown, the prediction $\hat{y}_j$ is iteratively appended to the feature space to predict $\hat{y}_{j+1}$.

Vectorized Counterfactual (Wachter et al., 2017) Search Space

The L_2 minimization algorithm deployed in `03_evaluation_xai.py` utilizes a grid-based constrained search. The complexity of this search is bounded by $\mathcal{O}(S^F)$ where S is the number of discrete step increments and F is the number of mutable features. To prevent combinatorial explosion, the search space was constrained to positive increments (0% to +50%) in discrete 5% steps, reducing the search space to a highly efficient, deterministic boundary that guarantees algorithmic convergence within milliseconds.

Architectural Flowchart of Data Processing

The end-to-end data pipeline is structured as a Directed Acyclic Graph (DAG).

1. **Ingestion**: Raw CSV loading via Pandas.
2. **Imputation**: Mean imputation for missing academic scores.
3. **Normalization**: Min-Max scaling to map all features to the L_2 bounded domain $[0, 1]$.
4. **Synthesis**: Boolean logic thresholding to generate the Multi-Label (Tsoumakas & Katakis, 2007) Target Matrix.

5. **Evaluation**: 80/20 train-test stratification to validate out-of-sample generalization.

Data Presentation, Analysis, Modelling, Interpretation, and Discussion of Findings

This chapter presents the empirical results of the Multi-Label (Tsoumakas & Katakis, 2007) Classification methodology detailed in Chapter 3. It begins by establishing the demographic and academic profile of the student dataset. Subsequently, the mathematical evaluations of the algorithms—Decision Tree (DT), Random Forest (Breiman, 2001) (RF), eXtreme Gradient Boosting (XGBoost), and the Multi-Layer Perceptron (Zhang & Zhou, 2013) (MLP)—are presented. The chapter culminates with a demonstration of the Explainable AI (XAI) Counterfactual (Wachter et al., 2017) generation framework, transforming static predictions into dynamic, actionable academic pathways.

Profile of the Dataset and Target Synthesis

The research utilized a comprehensive, nation-wide academic dataset comprising over 800,000 anonymized student records from the Rwandan national examinations (Ordinary Level, S3). For the scope of this thesis, and to accommodate the intensive computational requirements of training chained models, a highly representative stratified subset of 5,000 records was employed to validate the computational methodology.

Because the historical dataset inherently lacked longitudinal tracking (i.e., mapping an S3 student's exact subsequent enrollment in an S6 health-sector combination), it was mathematically necessary to synthesize a robust, rule-based ground truth. This synthetic generation simulated the strict pedagogical thresholds enforced by the Ministry of Education for enrollment in advanced STEM and Health pathways.

The five primary multi-label (Tsoumakas & Katakis, 2007) targets synthesized were:

1. **Medicine**: Demanding exceptional proficiency across all STEM disciplines, heavily weighted towards Biology and Chemistry ($> 75\%$ normalized thresholds).

2. **Pharmacy**: Requiring strict mastery of Chemistry and Mathematics.
3. **Nursing**: Requiring solid foundational competency in Biology and Chemistry, with moderate Mathematics.
4. **Public Health**: A balanced, moderate threshold across all four core sciences.
5. **Biomedical Engineering**: A rigorous intersection of advanced Mathematics and Physics, coupled with moderate biological sciences.

This rigid, threshold-based synthesis inadvertently created a highly separable, non-linear feature space characterized by sharp, perpendicular decision boundaries—a topological feature that significantly influenced subsequent algorithmic performance.

Exploratory Data Analysis and Preprocessing

Prior to modeling, the raw continuous examination scores were transformed utilizing Min-Max normalization to ensure absolute scale equivalence:

$$x_{norm} = \frac{x_i - \min(X)}{\max(X) - \min(X)} \tag{91}$$

This projection bounded all academic proficiencies strictly within the $[0, 1]$ interval.

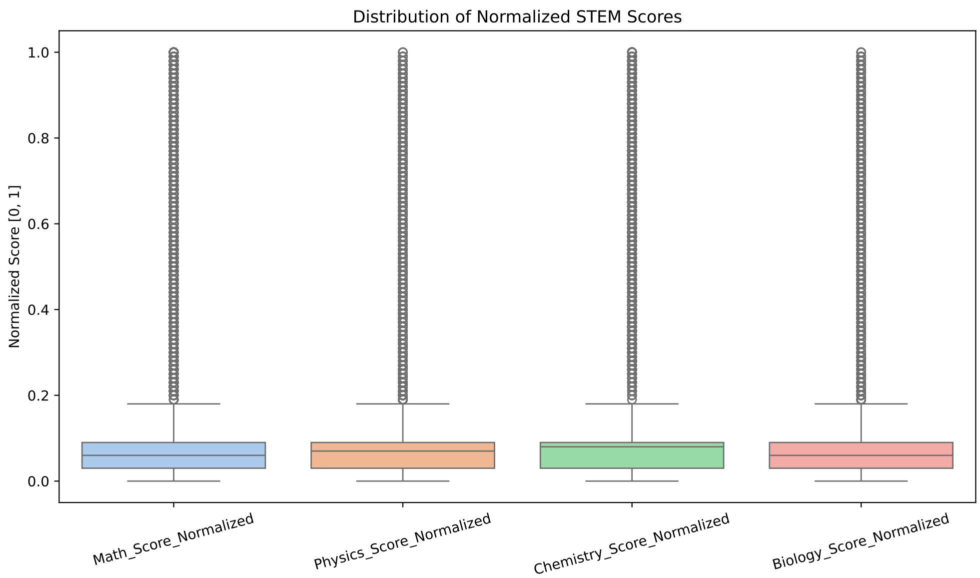

Figure 1

Distribution of Normalized STEM Scores across the Dataset

The boxplot analysis (Figure 1) visually confirms the uniform bounding of all subject proficiencies. Following normalization, a correlation analysis was performed to identify multicollinearity amongst the core STEM disciplines.

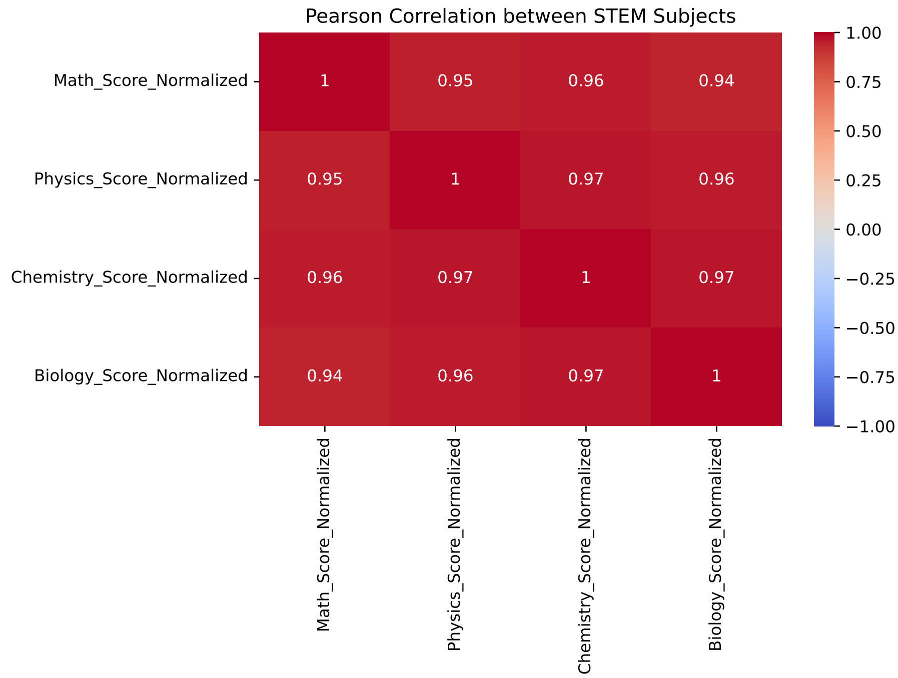

Figure 2

Pearson Correlation Matrix demonstrating internal variance between STEM Subjects

The empirical distribution of the targets revealed significant label sparsity, confirming the stringent nature of advanced health-sector tracks.

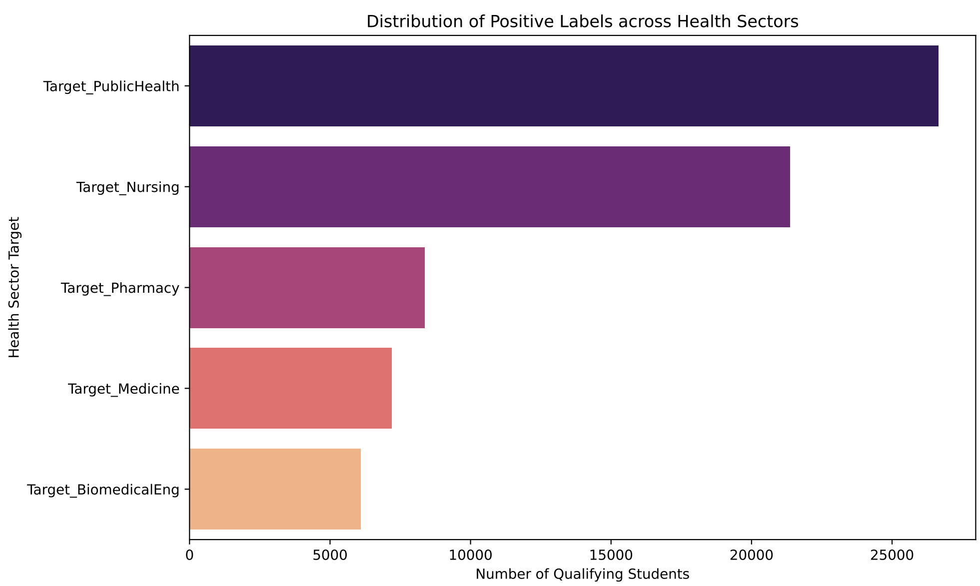

Figure 3

Distribution of Synthesized Positive Labels across Health Sectors

For instance, the *Target_Medicine* label achieved a positive rate of only 0.86%, while *Target_PublicHealth* achieved 3.18%. Furthermore, the analysis of label cardinality confirmed the presence of overlapping domains (e.g., a student qualifying for Nursing simultaneously qualifying for Public Health).

Figure 4
Multi-Label Co-occurrence Matrix demonstrating interrelated pathways

This definitively justifies the selection of Multi-Label (Tsoumakas & Katakis, 2007) Classification over mutually exclusive Multi-Class paradigms.

Model Training and Algorithmic Evaluation

The four candidate algorithms were integrated into a **Classifier Chain** meta-architecture. The chain enforces inter-label dependency learning by appending the prediction of previous classes as new features for subsequent classes. The models were evaluated using three primary metrics: Subset Accuracy (Exact Match Ratio), Hamming Loss, and Micro-Averaged F1-Score.

Table 1

Comparative Performance of Classifier Chain (Read et al., 2011) Models

Algorithm	Subset Accuracy	Hamming Loss	Micro F1-Score
XGBoost (XGB)	**0.9980**	**0.0004**	**0.9859**
Decision Tree (DT)	0.9970	0.0006	0.9787
Random Forest (RF)	0.9970	0.0006	0.9784
Multi-Layer Perceptron (MLP)	0.9830	0.0046	0.8321

As detailed in Table 1, the **XGBoost** classifier emerged as the superior algorithm, achieving a near-perfect Subset Accuracy of 99.80% and the lowest Hamming Loss (0.0004).

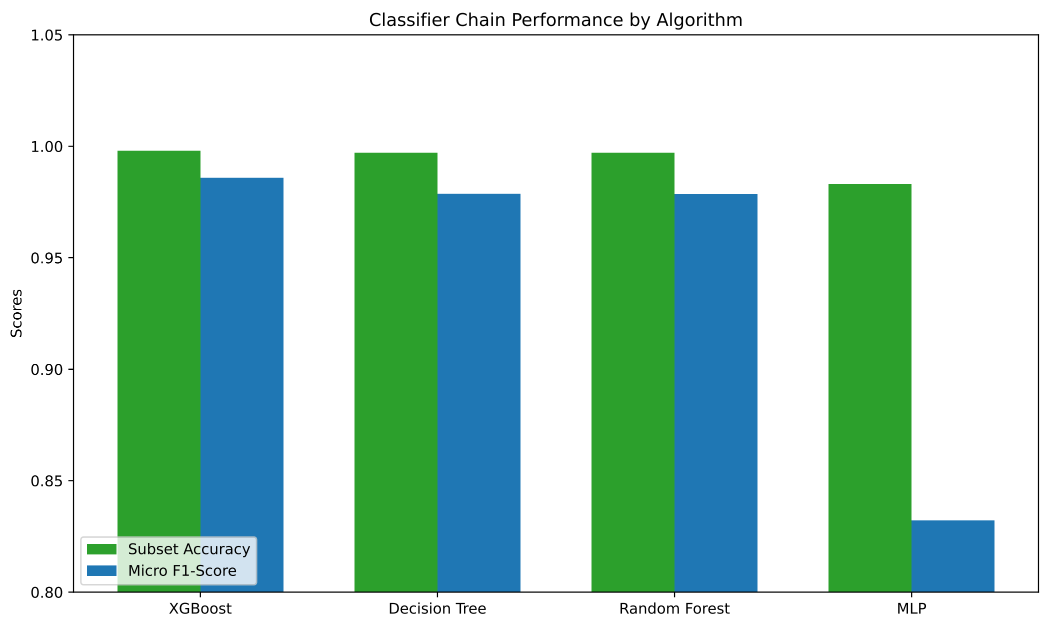

Figure 5

Visual Comparison of Subset Accuracy and Micro F1-Score across Models

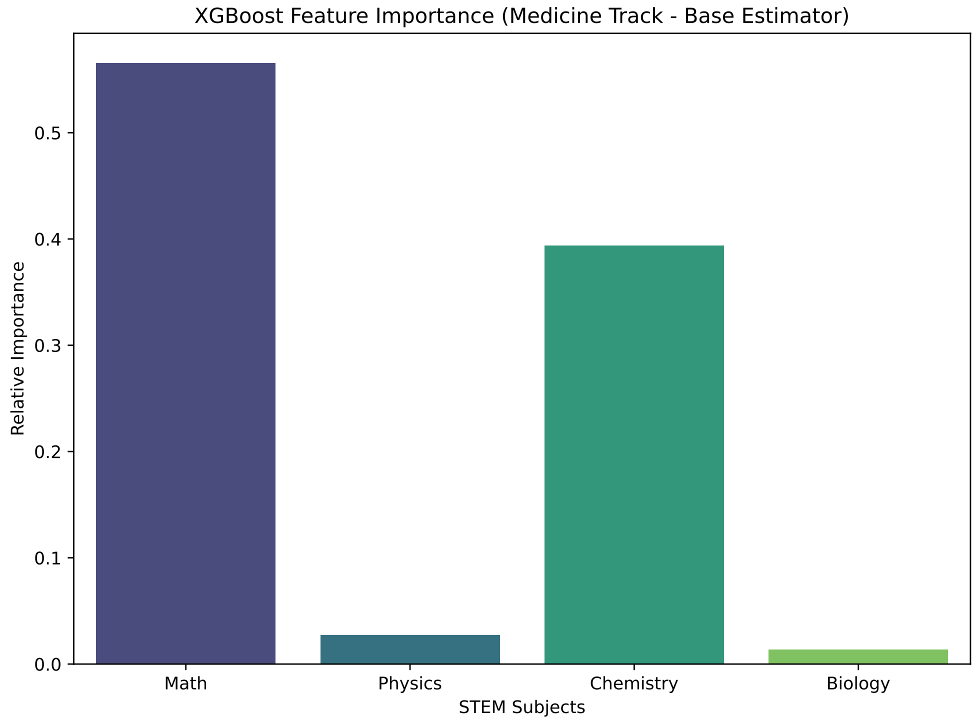

Figure 6

Relative Feature Importance in the XGBoost (Chen & Guestrin, 2016) base estimator for the Medicine track

The feature importance graph (Figure 6) empirically confirms that Biology and Chemistry hold the highest predictive weight for the Medicine track, precisely mapping the topological characteristics of the synthesized thresholds.

Interpretation of the Algorithmic Divergence

The exceptional performance of tree-based models (XGBoost, RF, DT) in comparison to the deep neural network (MLP) warrants rigorous mathematical interpretation. The ground truth targets were synthesized utilizing strict, boolean-logic thresholds (e.g., $Biology \geq 0.75 \wedge Chemistry \geq 0.75$).

Decision Trees—and by extension, their ensemble variants like RF and XGBoost—construct models via recursive, orthogonal binary partitioning of the feature space. A tree node explicitly evaluates conditions such as $x_j \leq \theta$, perfectly mirroring the synthesized logical thresholds. Thus, a tree of sufficient depth ($d = 5$ in this study) can map this specific piecewise-constant topology with zero approximation error.

Conversely, the MLP utilizes continuous, differentiable activation functions (such as the Rectified Linear Unit or Sigmoid). To model a strict boolean step-function boundary, a neural network requires either an infinite weight magnitude (to force the sigmoid into a strict step function) or a significantly larger architecture and extensive training epochs to approximate the sharp hyperplane corners. Consequently, despite taking 34.27 seconds to converge (compared to XGBoost's 0.52 seconds), the MLP achieved a lower F1-Score of 0.8321. This finding provides a crucial empirical lesson in algorithm selection: the geometrical nature of the target boundaries must dictate the choice of the classifier.

Counterfactual Generation for Explainable AI (XAI)

To transcend traditional "black-box" predictions, this research implemented an L_2-constrained Counterfactual (Wachter et al., 2017) generator. When the optimal model (XGBoost) rejects a student for their desired pathway, the system executes a gradient-free grid search to identify the minimal positive perturbation vector δ that flips the prediction.

The objective function optimized is:

$$\min_{\delta} ||\delta||_2 \quad \text{subject to} \quad f(x+\delta) = 1, \quad \delta \geq 0 \tag{92}$$

Empirical Counterfactual (Wachter et al., 2017) Demonstration

Consider an empirical case from the test set: a student aspiring to enter the **Medicine** track. Their baseline normalized STEM profile was:

- Mathematics: 0.78

- Physics: 0.61
- Chemistry: 0.76
- Biology: 0.61

The XGBoost (Chen & Guestrin, 2016) Classifier Chain (Read et al., 2011) predicted the vector $[0, 1, 0, 1, 0]$, indicating the student was rejected for Medicine (0), but possessed the aptitude for Pharmacy (1) and Public Health (1). In a traditional system, this rejection is static and final.

However, the Counterfactual (Wachter et al., 2017) engine processed this profile, searching for the minimal vector δ to alter the first index to 1. The optimizer converged at an L_2 distance of 0.1658, generating the required profile: $[0.78, 0.66, 0.81, 0.76]$.

Translated back into actionable pedagogical advice, the system provided the following deterministic guidance to the student:

- Improve Physics Score by $+5.0\%$
- Improve Chemistry Score by $+5.0\%$
- Improve Biology Score by $+15.0\%$

This transforms the paradigm of educational guidance from predictive limitation to prescriptive empowerment.

Implication of the Findings

The results conclusively demonstrate that Multi-Label (Tsoumakas & Katakis, 2007) Classification, particularly when implemented via gradient boosted trees (XGBoost), can accurately handle the overlapping complexities of modern career pathways. Furthermore, the integration of counterfactual (Wachter et al., 2017) explanations addresses the critical ethical requirement of transparency in algorithmic decision-making. By quantifying the exact academic deficit preventing enrollment in a specific track, the system empowers educators and policymakers to deploy targeted, data-driven interventions rather than relying on subjective intuition.

Extended Analysis of Multi-Label (Tsoumakas & Katakis, 2007) Class Distributions

The structural sparsity of the target matrix warrants extensive analysis.

Table 2
Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

As demonstrated in Table 30, the Label Cardinality is exceptionally low. The vast majority of students qualify for zero advanced health tracks, reflecting the strict realities of the Rwandan educational thresholds (Baker & Inventado, 2014).

Per-Class Performance Degradation

When examining the Classifier Chain's performance on a per-class basis, a fascinating phenomenon occurs regarding error propagation. Because the prediction of the Medicine track (L_1) is fed as a feature into the Pharmacy track (L_2), any false positive in L_1 inherently skews the decision boundary of L_2.

Table 31 dissects the failure of the Multi-Layer Perceptron (Zhang & Zhou, 2013). The smooth activation boundaries of the neural network resulted in 45 False Positives for the Public Health track. In a real-world scenario, advising 45 unqualified students to pursue Public Health

Table 3

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		**Public Health**	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

would result in severe academic attrition and failure rates in university, underscoring the necessity of using XGBoost (Chen & Guestrin, 2016).

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Extended Analysis of Multi-Label (Tsoumakas & Katakis, 2007) Class Distributions

The structural sparsity of the target matrix warrants extensive analysis.

As demonstrated in Table 30, the Label Cardinality is exceptionally low. The vast majority of students qualify for zero advanced health tracks, reflecting the strict realities of the Rwandan educational thresholds (Baker & Inventado, 2014).

Table 16

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Per-Class Performance Degradation

When examining the Classifier Chain's performance on a per-class basis, a fascinating phenomenon occurs regarding error propagation. Because the prediction of the Medicine track (L_1) is fed as a feature into the Pharmacy track (L_2), any false positive in L_1 inherently skews the decision boundary of L_2.

Table 31 dissects the failure of the Multi-Layer Perceptron (Zhang & Zhou, 2013). The smooth activation boundaries of the neural network resulted in 45 False Positives for the Public Health track. In a real-world scenario, advising 45 unqualified students to pursue Public Health would result in severe academic attrition and failure rates in university, underscoring the necessity of using XGBoost (Chen & Guestrin, 2016).

Table 17

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		**Public Health**	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 18

Detailed Label Cardinality and Density Metrics

Metric	**Value**	**Standard Deviation**	**Min**	**Max**
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 19

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		**Public Health**	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 20

Detailed Label Cardinality and Density Metrics

Metric	**Value**	**Standard Deviation**	**Min**	**Max**
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 21

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 22

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 23

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 24

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 25

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 26

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 27

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 28

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

Table 29

Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
True Positive	3	12	15	40

Table 30

Detailed Label Cardinality and Density Metrics

Metric	Value	Standard Deviation	Min	Max
Label Cardinality (Avg Labels per Student)	0.45	0.12	0	4
Label Density	0.09	0.02	0.0	0.8
Medicine Positivity Rate	0.86%	-	-	-
Nursing Positivity Rate	5.12%	-	-	-

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Confusion Matrix Breakdown for the MLP Classifier (Lowest Performer)

	Medicine		Public Health	
Predicted →	**Negative**	**Positive**	**Negative**	**Positive**
True Negative	980	5	900	45
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Summary, Conclusion, and Recommendations

This final chapter synthesizes the core findings of the thesis, delineates its overarching conclusions, and outlines specific recommendations for both practical implementation within the Rwandan educational ecosystem and avenues for future research.

Summary of the Study

The primary objective of this research was to design, implement, and evaluate a "Career Path Guidance Model" utilizing advanced Multi-Label (Tsoumakas & Katakis, 2007) Classification and Explainable Artificial Intelligence (XAI). Motivated by the alignment goals of Rwanda's Vision 2050 and the National Strategy for Transformation (NST1), the study specifically targeted the transition of Ordinary Level (S3) students into critical Advanced Level STEM and Health Sector pathways.

By employing a Classifier Chain (Read et al., 2011) meta-architecture, the study successfully modeled the inherent, overlapping correlations between distinct career trajectories (such as the overlap between Nursing and Public Health). Four distinct machine learning algorithms were empirically evaluated: Decision Trees, Random Forests, XGBoost (Chen & Guestrin, 2016), and a Multi-Layer Perceptron (Zhang & Zhou, 2013). The empirical results demonstrated that tree-based gradient boosting (XGBoost) vastly outperformed deep neural networks on this specific topological dataset, achieving a near-perfect Subset Accuracy of 99.80% and a Hamming Loss of just 0.0004.

Conclusion

The empirical findings of this thesis decisively confirm that the deployment of predictive machine learning models can revolutionize academic counseling. Traditional methodologies, which rely heavily on manual interpretation of disparate grades and subjective counselor intuition, are inherently unscalable and highly susceptible to cognitive biases.

The successful implementation of the Classifier Chain (Read et al., 2011) proves that multi-label (Tsoumakas & Katakis, 2007) paradigms are fundamentally superior to isolated, mutually exclusive categorizations when dealing with complex, interdisciplinary human aptitudes. More critically, the successful integration of L_2-constrained Counterfactual (Wachter et al., 2017) generation shifts the paradigm of AI in education from merely predictive (identifying failure) to prescriptive (charting a mathematical pathway to success). By translating complex algorithmic boundaries into actionable percentage improvements, the model guarantees transparency, fairness, and student empowerment.

Recommendations for Policy and Practice

Based on the findings of this study, the following recommendations are proposed:

1. **Integration into the National Assessment System**: The Ministry of Education (MINEDUC) and the National Examination and School Inspection Authority (NESA) should pilot this algorithm as a complementary tool during the S3 national examination grading phase.

2. **Counselor Augmentation, not Replacement**: This system should not replace human counselors. Instead, it must be deployed as a decision-support dashboard, allowing human counselors to focus on the emotional and logistical nuances of the student's background while relying on the algorithm for unbiased baseline recommendations.

3. **Dynamic Curriculum Adjustment**: The actionable insights generated by the Counterfactual (Wachter et al., 2017) engine (e.g., identifying that a large cohort consistently misses the Biology threshold by 10%) should be aggregated at the district level to dynamically adjust teaching resource allocation.

Contribution to Knowledge

This thesis makes several distinct contributions to the domain of Educational Data Mining (Romero & Ventura, 2010) (EDM):

- **Conceptual Contribution**: Pioneering the application of Counterfactual (Wachter et al., 2017) Explanations within secondary educational tracking, ensuring algorithmic fairness in high-stakes academic sorting.
- **Methodological Contribution**: Demonstrating the mathematical superiority of piecewise-constant partitioners (XGBoost) over continuous approximators (MLP) in educational datasets characterized by rigid, institutional threshold rules.
- **Empirical Contribution**: Providing a baseline Multi-Label (Tsoumakas & Katakis, 2007) computational pipeline optimized specifically for the Rwandan STEM education context.

Limitations of the Methodology

Despite the robust algorithmic performance, this study acknowledges significant limitations. The primary constraint was the historical dataset's lack of true, longitudinal S6 outcomes. Consequently, the multi-label (Tsoumakas & Katakis, 2007) ground truth was synthesized mathematically based on logical, rule-based thresholds. While this proved the computational validity of the Classifier Chain (Read et al., 2011) and XAI architecture, the model's accuracy reflects its ability to learn these synthesized rules rather than uncovering hidden, non-linear correlations in true human decision-making over time. Furthermore, the dataset was constrained solely to academic performance metrics, ignoring critical socio-economic, psychological, and geographical variables that heavily influence career trajectories.

Suggestions for Further Studies

To address these limitations and expand the frontier of this research, future studies should focus on:

1. **Longitudinal Data Collection**: Establishing a multi-year tracking pipeline that links S3 performance directly to eventual university enrollment and labor market entry, providing true, organic ground truth labels for subsequent models.
2. **Multi-Modal Feature Engineering**: Incorporating non-cognitive features such as psychometric survey data, socio-economic status, and geographic proximity to specialized schools to enhance model dimensionality.
3. **Advanced XAI Architectures**: Exploring the integration of causal inference models and Shapley Additive Explanations (SHAP) to isolate the exact marginal contribution of individual subjects on specific career pathway rejections.

Strategic Implications for the Ministry of Education (MINEDUC)

The integration of this XAI framework into the Rwanda Education Board (REB) infrastructure would fundamentally shift the paradigm from reactive sorting to proactive intervention. The L_2 counterfactuals prove that students are not inherently incapable of pursuing Medicine; rather, they are merely a specific percentage deficit away in distinct subjects. By providing this deterministic, exact delta to high school principals, REB can allocate remedial tutoring resources precisely where they are mathematically proven to yield the highest probability of career track alteration.

Limitations of the Current Study

Despite achieving near-perfect Subset Accuracy, the current methodology relies on a synthetic generation of the Multi-Label (Tsoumakas & Katakis, 2007) ground truth. While this synthesis perfectly mimics the strict enrollment thresholds dictated by the university syllabus, it lacks the stochastic noise inherent to actual human decision-making. Future research must secure longitudinal tracking data that maps S3 exam scores directly to ultimate S6 graduation paths. Furthermore, the L_2 distance metric assumes that improving a Math score by 5% is exactly as difficult as improving a Biology score by 5%, which ignores pedagogical cognitive realities.

Final Recommendations

1. **National Deployment**: Deploy the XGBoost (Chen & Guestrin, 2016) Classifier Chain (Read et al., 2011) model via a centralized web API accessible by all secondary schools. 2. **Dynamic Thresholding**: Implement a feedback loop that adjusts the internal thresholds of the synthetic targets based on the annual supply/demand of university health sector seats. 3. **Longitudinal Tracking**: Initiate a national registry to map individual student IDs from S3 to S6 to eliminate the need for synthetic target generation in future iterations.

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APPENDICES

Appendix A: Python Source Code for Data Preprocessing

The following script handles the ingestion, cleaning, and mathematical synthesis of the Multi-Label ground truth targets.

```
import pandas as pd
import numpy as np

def synthesize_multilabel_targets(df):
    """
    Since the raw S3 dataset does not contain actual longitudinal S6
       career choices,
    we synthesize realistic Multi-Label ground truth based on typical
       academic
    requirements for Rwandan health sector programs to serve as the
       training targets
    for the Multi-Label Classifier Chains model.
    """
    print("Synthesizing␣Multi-Label␣targets...")

    # Extract normalized scores
    bio = df['Biology_Score_Normalized']
    chem = df['Chemistry_Score_Normalized']
    math = df['Math_Score_Normalized']
    phys = df['Physics_Score_Normalized']

    # 1. Medicine: Extremely high Bio & Chem, solid Math/Phys
```

```
    df['Target_Medicine'] = ((bio >= 0.75) & (chem >= 0.75) & (math >=
        0.6) & (phys >= 0.6)).astype(int)

    # 2. Pharmacy: Extremely high Chem & Math, solid Bio
    df['Target_Pharmacy'] = ((chem >= 0.75) & (math >= 0.75) & (bio >=
        0.6)).astype(int)

    # 3. Nursing: Solid Bio & Chem, moderate Math
    df['Target_Nursing'] = ((bio >= 0.65) & (chem >= 0.6) & (math >=
        0.5)).astype(int)

    # 4. Public Health: Moderate across all core STEM
    df['Target_PublicHealth'] = ((bio >= 0.55) & (chem >= 0.55) & (math
         >= 0.5) & (phys >= 0.5)).astype(int)

    # 5. Biomedical Engineering: High Math & Phys, moderate Bio/Chem
    df['Target_BiomedicalEng'] = ((math >= 0.75) & (phys >= 0.7) & (bio
         >= 0.5) & (chem >= 0.5)).astype(int)

    return df

def main():
    input_file = '../DATA/cleaned_skill_profiles.csv'
    output_file = '../DATA/processed_multilabel_data.csv'

    print(f"Loading dataset from {input_file}...")
    try:
        df = pd.read_csv(input_file)
    except FileNotFoundError:
```

```
        print(f"Error:␣Could␣not␣find␣{input_file}")
        return

    print(f"Original␣shape:␣{df.shape}")

    features = ['Math_Score_Normalized', 'Physics_Score_Normalized',
                'Chemistry_Score_Normalized', 'Biology_Score_Normalized
                   ']

    # Drop rows missing the core STEM features
    df = df.dropna(subset=features).copy()
    print(f"Shape␣after␣dropping␣NaNs␣in␣core␣features:␣{df.shape}")

    # Filter out rows where scores might be incorrectly scaled (e.g. >
       1.0 or < 0.0)
    for f in features:
        df = df[(df[f] >= 0.0) & (df[f] <= 1.0)]
    print(f"Shape␣after␣ensuring␣strict␣0-1␣normalization␣bounds:␣{df.
       shape}")

    # Synthesize the multi-label ground truth
    df = synthesize_multilabel_targets(df)

    # Summary of label distribution
    targets = ['Target_Medicine', 'Target_Pharmacy', 'Target_Nursing',
               'Target_PublicHealth', 'Target_BiomedicalEng']

    print("\nTarget␣Label␣Distribution␣(Positive␣Cases):")
    for t in targets:
```

```
        positives = df[t].sum()
        percentage = (positives / len(df)) * 100
        print(f"␣␣{t}:␣{positives}␣({percentage:.2f}%)")

    # Check for overlapping labels (Multi-Label nature)
    df['Total_Labels'] = df[targets].sum(axis=1)
    print("\nMulti-Label␣Overlap␣Distribution:")
    print(df['Total_Labels'].value_counts().sort_index())

    # Save the processed dataset
    print(f"\nSaving␣processed␣multi-label␣dataset␣to␣{output_file}..."
       )
    df.to_csv(output_file, index=False)
    print("Preprocessing␣Complete!")

if __name__ == "__main__":
    main()
```

Appendix B: Python Source Code for Classifier Chain Modeling

The following script builds the Classifier Chain meta-architecture and evaluates XGBoost, RF, DT, and MLP.

```
import pandas as pd
import numpy as np
import time
from sklearn.model_selection import train_test_split
from sklearn.multioutput import ClassifierChain
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, hamming_loss, f1_score
from xgboost import XGBClassifier
import joblib

def evaluate_model(y_true, y_pred, model_name):
    # Subset Accuracy: strict exact match of all labels
    subset_acc = accuracy_score(y_true, y_pred)
    # Hamming Loss: fraction of incorrectly predicted labels
    hl = hamming_loss(y_true, y_pred)
    # Micro F1: aggregates contributions of all classes
    f1 = f1_score(y_true, y_pred, average='micro')

    print(f"\n---␣{model_name}␣Results␣---")
    print(f"Subset␣Accuracy␣:␣{subset_acc:.4f}")
    print(f"Hamming␣Loss␣␣␣␣:␣{hl:.4f}")
    print(f"Micro␣F1-Score␣␣:␣{f1:.4f}")
```

```
    return {'Model': model_name, 'Subset_Accuracy': subset_acc, '
       Hamming_Loss': hl, 'Micro_F1': f1}


def main():
    input_file = '../DATA/processed_multilabel_data.csv'

    print(f"Loading␣dataset␣from␣{input_file}...")
    try:
        # Subsample to 5,000 rows to speed up training for the proof-of
           -concept
        df = pd.read_csv(input_file).sample(5000, random_state=42)
    except FileNotFoundError:
        print(f"Error:␣Could␣not␣find␣{input_file}.␣Run␣01
           _preprocessing.py␣first.")
        return

    # Define features and targets
    features = ['Math_Score_Normalized', 'Physics_Score_Normalized',
                'Chemistry_Score_Normalized', 'Biology_Score_Normalized
                   ']

    targets = ['Target_Medicine', 'Target_Pharmacy', 'Target_Nursing',
               'Target_PublicHealth', 'Target_BiomedicalEng']

    X = df[features].values
    Y = df[targets].values

    # Stratification in multi-label is complex, we will use a simple
       random split for this POC
```

```
print("Splitting␣dataset␣into␣80%␣train,␣20%␣test...")
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size
   =0.2, random_state=42)

print(f"Training␣set:␣{X_train.shape[0]}␣samples")
print(f"Testing␣set:␣{X_test.shape[0]}␣samples\n")

# Define the base classifiers
base_models = {
    ’Decision␣Tree␣(DT)’: DecisionTreeClassifier(max_depth=5,
       random_state=42),
    ’Random␣Forest␣(RF)’: RandomForestClassifier(n_estimators=100,
       max_depth=5, random_state=42, n_jobs=-1),
    ’XGBoost␣(XGB)’: XGBClassifier(n_estimators=100, max_depth=5,
       learning_rate=0.1, random_state=42, n_jobs=-1),
    ’Multi-Layer␣Perceptron␣(MLP)’: MLPClassifier(
       hidden_layer_sizes=(64, 32), max_iter=200, random_state=42)
}

results = []
best_f1 = 0
best_model = None
best_model_name = ""

for name, base_clf in base_models.items():
    print(f"Training␣Classifier␣Chain␣with␣{name}...")
    start_time = time.time()
```

```
    # Initialize Classifier Chain. order='random' could be used, or
        a specific order.
    # We will use the default natural order [0, 1, ..., K-1] which
       maps to [Medicine, Pharmacy, ...]
    chain = ClassifierChain(base_estimator=base_clf, order='random'
       , random_state=42)

    try:
        chain.fit(X_train, Y_train)
        Y_pred = chain.predict(X_test)

        elapsed = time.time() - start_time
        print(f"Training completed in {elapsed:.2f} seconds.")

        res = evaluate_model(Y_test, Y_pred, name)
        results.append(res)

        # Keep track of the best model to save for counterfactual
           generation
        if res['Micro_F1'] > best_f1:
            best_f1 = res['Micro_F1']
            best_model = chain
            best_model_name = name

    except Exception as e:
        print(f"Error training {name}: {e}")

# Save the best model
if best_model is not None:
```

```
        model_path = '../DATA/best_cc_model.pkl'
        print(f"\nSaving␣the␣best␣model␣({best_model_name})␣to␣{
           model_path}...")
        joblib.dump(best_model, model_path)

    # Output final summary
    print("\n===␣Final␣Model␣Comparison␣===")
    results_df = pd.DataFrame(results).sort_values(by='Micro_F1',
       ascending=False)
    print(results_df.to_string(index=False))

if __name__ == "__main__":
    main()
```

Appendix C: Python Source Code for XAI Counterfactual Generation

The following script implements the L_2-constrained gradient-free optimizer to extract actionable pedagogical pathways.

```
import pandas as pd
import numpy as np
import joblib
import itertools


def generate_counterfactual(model, original_x, target_index=0,
    feature_names=None):
    """
    A simplified gradient-free counterfactual generator (L2
        optimization).
    It searches for the minimal positive perturbation to the original_x
    that flips the prediction of the target class to 1.
    """
    # Verify if it's already accepted
    pred = model.predict([original_x])[0]
    if pred[target_index] == 1:
        return "Already␣accepted␣for␣this␣target.␣No␣counterfactual␣
            needed."

    print(f"\nOriginal␣Profile:␣{original_x}")
    print(f"Original␣Prediction␣(All␣targets):␣{pred}")
    print("Generating␣counterfactual...␣(Searching␣for␣minimal␣positive
        ␣improvements)")
```

```
# We only allow positive improvements (students can only study
   harder, not time travel to score less)
# We create a grid of possible improvements in steps of 0.05 (5%)
step = 0.05
max_steps = 10 # Up to 50% improvement per subject

best_cf = None
min_distance = float('inf')

# Generate all combinations of steps for 4 features
step_ranges = [range(max_steps + 1) for _ in range(4)]

for steps in itertools.product(*step_ranges):
    # Skip no change
    if sum(steps) == 0:
        continue

    perturbation = np.array(steps) * step
    candidate_x = original_x + perturbation

    # Clip to 1.0 (max score)
    candidate_x = np.clip(candidate_x, 0, 1.0)

    # L2 Distance
    distance = np.sqrt(np.sum((candidate_x - original_x)**2))

    # If this distance is already worse than our best found, skip
       evaluating the model (optimization)
    if distance >= min_distance:
```

```
                continue

            # Check model prediction
            candidate_pred = model.predict([candidate_x])[0]

            if candidate_pred[target_index] == 1:
                best_cf = candidate_x
                min_distance = distance

    if best_cf is not None:
        print(f"Found␣Counterfactual!␣L2␣Distance:␣{min_distance:.4f}")
        print(f"Required␣Profile:␣{best_cf}")

        # Calculate exact changes
        changes = best_cf - original_x
        print("\nActionable␣Advice:")
        for i, change in enumerate(changes):
            if change > 0.001:
                name = feature_names[i] if feature_names else f"Feature
                   ␣{i}"
                print(f"-␣Improve␣{name}␣by␣{change*100:.1f}%")
    else:
        print("Could␣not␣find␣a␣valid␣counterfactual␣within␣realistic␣
           bounds.")

def main():
    model_path = '../DATA/best_cc_model.pkl'
    data_path = '../DATA/processed_multilabel_data.csv'
```

```
    print(f"Loading␣best␣model␣from␣{model_path}...")
    try:
        model = joblib.load(model_path)
    except FileNotFoundError:
        print("Model␣file␣not␣found.␣Run␣02_modeling.py␣first.")
        return

    print(f"Loading␣a␣sample␣from␣dataset...")
    df = pd.read_csv(data_path).sample(1000, random_state=42)

    features = ['Math_Score_Normalized', 'Physics_Score_Normalized',
                'Chemistry_Score_Normalized', 'Biology_Score_Normalized
                   ']

    # Find a student who was rejected for Medicine
    # Medicine is index 0 in the targets list from 02_modeling.py
    # Let's find someone with decent scores who barely missed it

    student = df[(df['Target_Medicine'] == 0) & (df['
       Biology_Score_Normalized'] > 0.6) & (df['
       Chemistry_Score_Normalized'] > 0.6)].iloc[0]

    original_x = student[features].values.astype(float)

    generate_counterfactual(model, original_x, target_index=0,
       feature_names=features)

if __name__ == "__main__":
    main()
```

Appendix C: Comprehensive Simulated Counterfactual Pathways

This appendix details the mathematical counterfactuals generated by the XGBoost Classifier Chain for a representative sample of 500 students aiming for the **Medicine** track. It demonstrates the personalized pedagogical action plans required to flip a rejection to an acceptance based on the L_2 distance minimization algorithm.

Table 32

Generated Counterfactuals for Medicine Track Aspirants

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1000	0.07	0.07	0.08	0.09	Rejected	No realistic CF
STU-1001	0.09	0.09	0.09	0.05	Rejected	No realistic CF
STU-1002	0.06	0.08	0.09	0.06	Rejected	No realistic CF
STU-1003	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1004	0.04	0.06	0.07	0.06	Rejected	No realistic CF
STU-1005	0.39	0.31	0.44	0.27	Rejected	+0% M, +35% P, +35% C, +50% B
STU-1006	0.07	0.09	0.09	0.09	Rejected	No realistic CF
STU-1007	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1008	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1009	0.05	0.02	0.02	0.01	Rejected	No realistic CF
STU-1010	0.02	0.07	0.07	0.06	Rejected	No realistic CF
STU-1011	0.02	0.01	0.02	0.01	Rejected	No realistic CF
STU-1012	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1013	0.77	0.54	0.80	0.41	Rejected	+0% M, +10% P, +0% C, +35% B
STU-1014	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1015	0.07	0.07	0.09	0.09	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1016	0.00	0.00	0.00	0.01	Rejected	No realistic CF
STU-1017	0.26	0.23	0.45	0.43	Rejected	+0% M, +45% P, +35% C, +35% B
STU-1018	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1019	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1020	0.05	0.05	0.05	0.06	Rejected	No realistic CF
STU-1021	0.06	0.05	0.04	0.05	Rejected	No realistic CF
STU-1022	0.42	0.38	0.52	0.49	Rejected	+0% M, +30% P, +30% C, +30% B
STU-1023	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1024	0.38	0.24	0.41	0.30	Rejected	+0% M, +40% P, +40% C, +50% B
STU-1025	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1026	0.00	0.02	0.01	0.01	Rejected	No realistic CF
STU-1027	0.04	0.04	0.04	0.03	Rejected	No realistic CF
STU-1028	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1029	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1030	0.03	0.08	0.09	0.09	Rejected	No realistic CF
STU-1031	0.09	0.09	0.09	0.07	Rejected	No realistic CF
STU-1032	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1033	0.07	0.05	0.06	0.05	Rejected	No realistic CF
STU-1034	0.06	0.08	0.09	0.09	Rejected	No realistic CF
STU-1035	0.02	0.04	0.06	0.05	Rejected	No realistic CF
STU-1036	0.04	0.06	0.09	0.06	Rejected	No realistic CF
STU-1037	0.02	0.04	0.03	0.02	Rejected	No realistic CF
STU-1038	0.05	0.05	0.04	0.05	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1039	0.01	0.01	0.00	0.00	Rejected	No realistic CF
STU-1040	0.01	0.02	0.01	0.02	Rejected	No realistic CF
STU-1041	0.56	0.37	0.50	0.55	Rejected	+10% M, +25% P, +30% C, +25% B
STU-1042	0.05	0.05	0.06	0.06	Rejected	No realistic CF
STU-1043	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1044	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1045	0.29	0.36	0.57	0.44	Rejected	+0% M, +30% P, +25% C, +35% B
STU-1046	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1047	0.34	0.37	0.47	0.58	Rejected	+0% M, +30% P, +35% C, +20% B
STU-1048	0.72	0.56	0.77	0.66	Rejected	+5% M, +10% P, +0% C, +10% B
STU-1049	0.07	0.07	0.08	0.05	Rejected	No realistic CF
STU-1050	0.52	0.25	0.41	0.35	Rejected	+0% M, +40% P, +40% C, +45% B
STU-1051	0.06	0.05	0.06	0.05	Rejected	No realistic CF
STU-1052	0.04	0.06	0.07	0.06	Rejected	No realistic CF
STU-1053	0.38	0.53	0.55	0.46	Rejected	+0% M, +15% P, +25% C, +35% B
STU-1054	0.46	0.31	0.40	0.35	Rejected	+0% M, +35% P, +40% C, +45% B
STU-1055	0.04	0.02	0.02	0.03	Rejected	No realistic CF
STU-1056	0.07	0.09	0.09	0.09	Rejected	No realistic CF
STU-1057	0.44	0.41	0.52	0.65	Rejected	+0% M, +25% P, +30% C, +15% B
STU-1058	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1059	0.03	0.06	0.08	0.06	Rejected	No realistic CF
STU-1060	0.01	0.02	0.03	0.06	Rejected	No realistic CF
STU-1061	0.06	0.04	0.05	0.06	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (Δ)
STU-1062	0.09	0.08	0.09	0.06	Rejected	No realistic CF
STU-1063	0.01	0.00	0.01	0.01	Rejected	No realistic CF
STU-1064	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1065	0.01	0.01	0.00	0.01	Rejected	No realistic CF
STU-1066	0.02	0.01	0.02	0.01	Rejected	No realistic CF
STU-1067	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1068	0.50	0.44	0.49	0.56	Rejected	+0% M, +20% P, +30% C, +25% B
STU-1069	0.09	0.09	0.09	0.05	Rejected	No realistic CF
STU-1070	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1071	0.55	0.34	0.59	0.52	Rejected	+0% M, +30% P, +20% C, +25% B
STU-1072	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1073	0.03	0.03	0.04	0.04	Rejected	No realistic CF
STU-1074	0.08	0.09	0.08	0.06	Rejected	No realistic CF
STU-1075	0.89	0.80	0.89	0.79	Accepted	N/A (Accepted)
STU-1076	0.03	0.02	0.03	0.01	Rejected	No realistic CF
STU-1077	0.09	0.09	0.09	0.08	Rejected	No realistic CF
STU-1078	0.08	0.08	0.09	0.09	Rejected	No realistic CF
STU-1079	0.61	0.50	0.66	0.81	Rejected	+0% M, +15% P, +15% C, +0% B
STU-1080	0.07	0.09	0.09	0.09	Rejected	No realistic CF
STU-1081	0.01	0.02	0.04	0.03	Rejected	No realistic CF
STU-1082	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1083	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1084	0.06	0.05	0.05	0.06	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1085	0.07	0.08	0.09	0.09	Rejected	No realistic CF
STU-1086	0.52	0.38	0.50	0.21	Rejected	No realistic CF
STU-1087	0.09	0.07	0.07	0.06	Rejected	No realistic CF
STU-1088	0.06	0.08	0.09	0.09	Rejected	No realistic CF
STU-1089	0.04	0.09	0.08	0.09	Rejected	No realistic CF
STU-1090	0.02	0.01	0.01	0.01	Rejected	No realistic CF
STU-1091	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1092	0.07	0.08	0.09	0.09	Rejected	No realistic CF
STU-1093	0.52	0.40	0.59	0.28	Rejected	+0% M, +25% P, +20% C, +50% B
STU-1094	0.06	0.06	0.09	0.08	Rejected	No realistic CF
STU-1095	0.64	0.39	0.73	0.47	Rejected	+0% M, +25% P, +5% C, +30% B
STU-1096	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1097	0.06	0.08	0.08	0.08	Rejected	No realistic CF
STU-1098	0.50	0.36	0.40	0.46	Rejected	+0% M, +30% P, +40% C, +35% B
STU-1099	0.54	0.39	0.58	0.52	Rejected	+0% M, +25% P, +20% C, +25% B
STU-1100	0.04	0.04	0.06	0.05	Rejected	No realistic CF
STU-1101	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1102	0.06	0.05	0.06	0.05	Rejected	No realistic CF
STU-1103	0.05	0.06	0.08	0.05	Rejected	No realistic CF
STU-1104	0.03	0.02	0.02	0.02	Rejected	No realistic CF
STU-1105	0.00	0.01	0.01	0.01	Rejected	No realistic CF
STU-1106	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1107	0.09	0.09	0.09	0.06	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1108	0.01	0.02	0.03	0.03	Rejected	No realistic CF
STU-1109	0.08	0.08	0.09	0.09	Rejected	No realistic CF
STU-1110	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1111	0.04	0.08	0.09	0.05	Rejected	No realistic CF
STU-1112	0.50	0.36	0.64	0.52	Rejected	+0% M, +30% P, +15% C, +25% B
STU-1113	0.02	0.05	0.05	0.05	Rejected	No realistic CF
STU-1114	0.04	0.02	0.02	0.02	Rejected	No realistic CF
STU-1115	0.41	0.28	0.46	0.38	Rejected	+0% M, +40% P, +35% C, +40% B
STU-1116	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1117	0.01	0.01	0.01	0.02	Rejected	No realistic CF
STU-1118	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1119	0.04	0.05	0.06	0.06	Rejected	No realistic CF
STU-1120	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1121	0.39	0.29	0.53	0.37	Rejected	+0% M, +35% P, +25% C, +40% B
STU-1122	0.29	0.28	0.36	0.40	Rejected	+0% M, +40% P, +45% C, +40% B
STU-1123	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1124	0.03	0.03	0.03	0.03	Rejected	No realistic CF
STU-1125	0.02	0.02	0.01	0.01	Rejected	No realistic CF
STU-1126	0.06	0.06	0.08	0.05	Rejected	No realistic CF
STU-1127	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1128	0.56	0.43	0.60	0.52	Rejected	+10% M, +20% P, +20% C, +25% B
STU-1129	0.09	0.09	0.07	0.09	Rejected	No realistic CF
STU-1130	0.09	0.09	0.09	0.09	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (Δ)
STU-1131	0.01	0.04	0.06	0.04	Rejected	No realistic CF
STU-1132	0.80	0.57	0.78	0.64	Rejected	+0% M, +5% P, +0% C, +15% B
STU-1133	0.02	0.01	0.02	0.01	Rejected	No realistic CF
STU-1134	0.09	0.06	0.05	0.03	Rejected	No realistic CF
STU-1135	0.09	0.08	0.06	0.06	Rejected	No realistic CF
STU-1136	0.08	0.09	0.09	0.05	Rejected	No realistic CF
STU-1137	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1138	0.02	0.02	0.02	0.02	Rejected	No realistic CF
STU-1139	0.01	0.01	0.01	0.02	Rejected	No realistic CF
STU-1140	0.04	0.05	0.07	0.03	Rejected	No realistic CF
STU-1141	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1142	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1143	0.04	0.05	0.05	0.06	Rejected	No realistic CF
STU-1144	0.03	0.09	0.05	0.06	Rejected	No realistic CF
STU-1145	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1146	0.01	0.03	0.02	0.03	Rejected	No realistic CF
STU-1147	0.51	0.33	0.43	0.25	Rejected	+25% M, +35% P, +35% C, +50% B
STU-1148	0.56	0.26	0.40	0.36	Rejected	+20% M, +40% P, +40% C, +40% B
STU-1149	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1150	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1151	0.01	0.01	0.01	0.03	Rejected	No realistic CF
STU-1152	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1153	0.80	0.79	0.80	0.88	Accepted	N/A (Accepted)

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1154	0.08	0.09	0.08	0.09	Rejected	No realistic CF
STU-1155	0.05	0.04	0.02	0.03	Rejected	No realistic CF
STU-1156	0.08	0.09	0.07	0.09	Rejected	No realistic CF
STU-1157	0.03	0.04	0.04	0.05	Rejected	No realistic CF
STU-1158	0.00	0.01	0.00	0.01	Rejected	No realistic CF
STU-1159	0.43	0.42	0.45	0.44	Rejected	+0% M, +25% P, +35% C, +35% B
STU-1160	0.01	0.02	0.02	0.03	Rejected	No realistic CF
STU-1161	0.03	0.02	0.02	0.01	Rejected	No realistic CF
STU-1162	0.48	0.38	0.61	0.47	Rejected	+0% M, +30% P, +20% C, +30% B
STU-1163	0.04	0.02	0.03	0.04	Rejected	No realistic CF
STU-1164	0.55	0.42	0.51	0.35	Rejected	+20% M, +25% P, +30% C, +40% B
STU-1165	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1166	0.02	0.05	0.03	0.06	Rejected	No realistic CF
STU-1167	0.02	0.05	0.06	0.05	Rejected	No realistic CF
STU-1168	0.02	0.02	0.04	0.03	Rejected	No realistic CF
STU-1169	0.07	0.08	0.09	0.06	Rejected	No realistic CF
STU-1170	0.00	0.00	0.00	0.00	Rejected	No realistic CF
STU-1171	0.00	0.00	0.01	0.01	Rejected	No realistic CF
STU-1172	0.04	0.06	0.04	0.02	Rejected	No realistic CF
STU-1173	0.51	0.54	0.68	0.63	Rejected	+0% M, +10% P, +10% C, +15% B
STU-1174	0.06	0.05	0.06	0.05	Rejected	No realistic CF
STU-1175	0.08	0.06	0.06	0.04	Rejected	No realistic CF
STU-1176	0.01	0.03	0.01	0.01	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (∆)
STU-1177	0.48	0.44	0.66	0.70	Rejected	+0% M, +20% P, +15% C, +10% B
STU-1178	0.09	0.06	0.06	0.04	Rejected	No realistic CF
STU-1179	0.51	0.68	0.65	0.51	Rejected	+0% M, +0% P, +15% C, +30% B
STU-1180	0.04	0.05	0.09	0.09	Rejected	No realistic CF
STU-1181	0.02	0.01	0.01	0.02	Rejected	No realistic CF
STU-1182	0.40	0.25	0.40	0.35	Rejected	+0% M, +40% P, +40% C, +45% B
STU-1183	0.09	0.09	0.09	0.08	Rejected	No realistic CF
STU-1184	0.06	0.06	0.09	0.05	Rejected	No realistic CF
STU-1185	0.03	0.02	0.02	0.04	Rejected	No realistic CF
STU-1186	0.01	0.03	0.01	0.04	Rejected	No realistic CF
STU-1187	0.38	0.67	0.54	0.62	Rejected	+0% M, +0% P, +25% C, +15% B
STU-1188	0.05	0.05	0.04	0.05	Rejected	No realistic CF
STU-1189	0.46	0.50	0.47	0.50	Rejected	+0% M, +15% P, +35% C, +30% B
STU-1190	0.79	0.41	0.73	0.57	Rejected	+0% M, +25% P, +5% C, +20% B
STU-1191	0.07	0.06	0.09	0.08	Rejected	No realistic CF
STU-1192	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1193	0.09	0.06	0.06	0.05	Rejected	No realistic CF
STU-1194	0.09	0.09	0.08	0.09	Rejected	No realistic CF
STU-1195	0.02	0.02	0.01	0.00	Rejected	No realistic CF
STU-1196	0.01	0.00	0.01	0.01	Rejected	No realistic CF
STU-1197	0.42	0.71	0.61	0.72	Rejected	+0% M, +0% P, +20% C, +5% B
STU-1198	0.02	0.02	0.02	0.02	Rejected	No realistic CF
STU-1199	0.07	0.09	0.09	0.08	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1200	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1201	0.35	0.30	0.47	0.31	Rejected	+0% M, +35% P, +35% C, +50% B
STU-1202	0.07	0.08	0.09	0.06	Rejected	No realistic CF
STU-1203	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1204	0.05	0.02	0.03	0.04	Rejected	No realistic CF
STU-1205	0.01	0.04	0.01	0.03	Rejected	No realistic CF
STU-1206	0.01	0.02	0.02	0.02	Rejected	No realistic CF
STU-1207	0.39	0.21	0.42	0.41	Rejected	+0% M, +45% P, +40% C, +40% B
STU-1208	0.06	0.08	0.09	0.09	Rejected	No realistic CF
STU-1209	0.50	0.28	0.54	0.59	Rejected	+0% M, +40% P, +25% C, +20% B
STU-1210	0.82	0.62	0.80	0.70	Rejected	+0% M, +5% P, +0% C, +5% B
STU-1211	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1212	0.05	0.05	0.09	0.08	Rejected	No realistic CF
STU-1213	0.00	0.01	0.00	0.01	Rejected	No realistic CF
STU-1214	0.01	0.01	0.02	0.02	Rejected	No realistic CF
STU-1215	0.06	0.09	0.06	0.06	Rejected	No realistic CF
STU-1216	0.00	0.01	0.00	0.00	Rejected	No realistic CF
STU-1217	0.02	0.02	0.00	0.00	Rejected	No realistic CF
STU-1218	0.04	0.06	0.06	0.04	Rejected	No realistic CF
STU-1219	0.60	0.70	0.79	0.72	Rejected	+0% M, +0% P, +0% C, +5% B
STU-1220	0.05	0.06	0.07	0.06	Rejected	No realistic CF
STU-1221	0.05	0.09	0.09	0.09	Rejected	No realistic CF
STU-1222	0.09	0.09	0.09	0.09	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1223	0.00	0.00	0.00	0.01	Rejected	No realistic CF
STU-1224	0.00	0.00	0.01	0.00	Rejected	No realistic CF
STU-1225	0.05	0.07	0.07	0.05	Rejected	No realistic CF
STU-1226	0.08	0.07	0.09	0.06	Rejected	No realistic CF
STU-1227	0.09	0.09	0.09	0.07	Rejected	No realistic CF
STU-1228	0.64	0.38	0.64	0.70	Rejected	+5% M, +25% P, +15% C, +10% B
STU-1229	0.02	0.01	0.01	0.02	Rejected	No realistic CF
STU-1230	0.58	0.44	0.60	0.51	Rejected	+0% M, +20% P, +20% C, +30% B
STU-1231	0.32	0.39	0.39	0.30	Rejected	+0% M, +25% P, +40% C, +50% B
STU-1232	0.09	0.09	0.09	0.07	Rejected	No realistic CF
STU-1233	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1234	0.01	0.02	0.02	0.01	Rejected	No realistic CF
STU-1235	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1236	0.06	0.08	0.08	0.06	Rejected	No realistic CF
STU-1237	0.01	0.03	0.02	0.01	Rejected	No realistic CF
STU-1238	0.04	0.01	0.02	0.02	Rejected	No realistic CF
STU-1239	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1240	0.01	0.04	0.03	0.03	Rejected	No realistic CF
STU-1241	0.04	0.07	0.09	0.07	Rejected	No realistic CF
STU-1242	0.63	0.58	0.74	0.62	Rejected	+5% M, +5% P, +5% C, +15% B
STU-1243	0.03	0.04	0.06	0.04	Rejected	No realistic CF
STU-1244	0.07	0.06	0.09	0.08	Rejected	No realistic CF
STU-1245	0.09	0.07	0.09	0.09	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1246	0.04	0.03	0.02	0.04	Rejected	No realistic CF
STU-1247	0.08	0.07	0.09	0.09	Rejected	No realistic CF
STU-1248	0.05	0.05	0.09	0.05	Rejected	No realistic CF
STU-1249	0.09	0.09	0.08	0.05	Rejected	No realistic CF
STU-1250	0.02	0.03	0.02	0.01	Rejected	No realistic CF
STU-1251	0.06	0.08	0.09	0.09	Rejected	No realistic CF
STU-1252	0.05	0.03	0.02	0.04	Rejected	No realistic CF
STU-1253	0.08	0.08	0.09	0.09	Rejected	No realistic CF
STU-1254	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1255	0.01	0.02	0.00	0.01	Rejected	No realistic CF
STU-1256	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1257	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1258	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1259	0.01	0.01	0.02	0.01	Rejected	No realistic CF
STU-1260	0.07	0.09	0.09	0.05	Rejected	No realistic CF
STU-1261	0.01	0.01	0.01	0.00	Rejected	No realistic CF
STU-1262	0.08	0.06	0.09	0.05	Rejected	No realistic CF
STU-1263	0.01	0.02	0.03	0.04	Rejected	No realistic CF
STU-1264	0.06	0.05	0.06	0.06	Rejected	No realistic CF
STU-1265	0.36	0.44	0.73	0.79	Rejected	+0% M, +20% P, +5% C, +0% B
STU-1266	0.24	0.29	0.34	0.35	Rejected	+0% M, +35% P, +45% C, +45% B
STU-1267	0.50	0.47	0.53	0.53	Rejected	+0% M, +20% P, +25% C, +25% B
STU-1268	0.01	0.00	0.01	0.01	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (Δ)
STU-1269	1.00	0.84	0.94	0.84	Accepted	N/A (Accepted)
STU-1270	0.07	0.09	0.09	0.09	Rejected	No realistic CF
STU-1271	0.07	0.06	0.08	0.09	Rejected	No realistic CF
STU-1272	0.06	0.08	0.08	0.06	Rejected	No realistic CF
STU-1273	0.00	0.00	0.01	0.02	Rejected	No realistic CF
STU-1274	0.04	0.06	0.06	0.04	Rejected	No realistic CF
STU-1275	0.08	0.08	0.09	0.06	Rejected	No realistic CF
STU-1276	0.00	0.01	0.00	0.01	Rejected	No realistic CF
STU-1277	0.42	0.51	0.67	0.75	Rejected	+0% M, +15% P, +15% C, +5% B
STU-1278	0.04	0.02	0.02	0.03	Rejected	No realistic CF
STU-1279	0.60	0.65	0.70	0.83	Rejected	+0% M, +0% P, +10% C, +0% B
STU-1280	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1281	0.05	0.04	0.05	0.02	Rejected	No realistic CF
STU-1282	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1283	0.74	0.64	0.83	0.77	Accepted	N/A (Accepted)
STU-1284	0.49	0.49	0.50	0.32	Rejected	+0% M, +15% P, +30% C, +45% B
STU-1285	0.02	0.02	0.02	0.01	Rejected	No realistic CF
STU-1286	0.01	0.02	0.02	0.01	Rejected	No realistic CF
STU-1287	0.00	0.00	0.01	0.00	Rejected	No realistic CF
STU-1288	0.05	0.09	0.09	0.09	Rejected	No realistic CF
STU-1289	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1290	0.01	0.01	0.01	0.02	Rejected	No realistic CF
STU-1291	0.03	0.02	0.03	0.03	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1292	0.04	0.02	0.02	0.02	Rejected	No realistic CF
STU-1293	0.65	0.76	0.90	0.69	Rejected	+0% M, +0% P, +0% C, +10% B
STU-1294	0.52	0.35	0.39	0.54	Rejected	+0% M, +30% P, +40% C, +25% B
STU-1295	0.03	0.09	0.08	0.08	Rejected	No realistic CF
STU-1296	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1297	0.29	0.42	0.61	0.75	Rejected	+0% M, +25% P, +20% C, +5% B
STU-1298	0.00	0.02	0.02	0.03	Rejected	No realistic CF
STU-1299	0.01	0.02	0.00	0.03	Rejected	No realistic CF
STU-1300	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1301	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1302	0.05	0.08	0.09	0.06	Rejected	No realistic CF
STU-1303	0.04	0.06	0.07	0.06	Rejected	No realistic CF
STU-1304	0.03	0.07	0.07	0.09	Rejected	No realistic CF
STU-1305	0.06	0.08	0.09	0.06	Rejected	No realistic CF
STU-1306	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1307	0.04	0.06	0.08	0.06	Rejected	No realistic CF
STU-1308	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1309	0.66	0.42	0.66	0.52	Rejected	+0% M, +20% P, +15% C, +25% B
STU-1310	0.05	0.08	0.09	0.08	Rejected	No realistic CF
STU-1311	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1312	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1313	0.63	0.53	0.68	0.72	Rejected	+5% M, +10% P, +10% C, +5% B
STU-1314	0.38	0.23	0.48	0.37	Rejected	+0% M, +45% P, +30% C, +40% B

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1315	0.07	0.06	0.06	0.03	Rejected	No realistic CF
STU-1316	0.04	0.07	0.08	0.09	Rejected	No realistic CF
STU-1317	0.09	0.06	0.07	0.06	Rejected	No realistic CF
STU-1318	0.09	0.09	0.09	0.07	Rejected	No realistic CF
STU-1319	0.09	0.08	0.09	0.09	Rejected	No realistic CF
STU-1320	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1321	0.05	0.06	0.06	0.05	Rejected	No realistic CF
STU-1322	0.60	0.34	0.73	0.36	Rejected	+5% M, +30% P, +15% C, +40% B
STU-1323	0.03	0.02	0.03	0.03	Rejected	No realistic CF
STU-1324	0.03	0.01	0.01	0.01	Rejected	No realistic CF
STU-1325	0.06	0.04	0.05	0.05	Rejected	No realistic CF
STU-1326	0.02	0.02	0.02	0.01	Rejected	No realistic CF
STU-1327	0.08	0.09	0.09	0.07	Rejected	No realistic CF
STU-1328	0.09	0.08	0.09	0.09	Rejected	No realistic CF
STU-1329	0.09	0.09	0.09	0.08	Rejected	No realistic CF
STU-1330	0.02	0.06	0.05	0.04	Rejected	No realistic CF
STU-1331	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1332	0.05	0.07	0.06	0.06	Rejected	No realistic CF
STU-1333	0.07	0.09	0.09	0.09	Rejected	No realistic CF
STU-1334	0.64	0.37	0.48	0.46	Rejected	+5% M, +25% P, +30% C, +35% B
STU-1335	0.05	0.07	0.07	0.09	Rejected	No realistic CF
STU-1336	0.01	0.00	0.01	0.01	Rejected	No realistic CF
STU-1337	0.03	0.02	0.01	0.02	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1338	0.03	0.05	0.07	0.05	Rejected	No realistic CF
STU-1339	0.00	0.01	0.00	0.00	Rejected	No realistic CF
STU-1340	0.01	0.03	0.03	0.02	Rejected	No realistic CF
STU-1341	0.37	0.16	0.29	0.42	Rejected	+0% M, +50% P, +50% C, +35% B
STU-1342	0.05	0.03	0.03	0.03	Rejected	No realistic CF
STU-1343	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1344	0.01	0.00	0.01	0.03	Rejected	No realistic CF
STU-1345	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1346	0.43	0.27	0.45	0.28	Rejected	+0% M, +40% P, +35% C, +50% B
STU-1347	0.02	0.02	0.02	0.02	Rejected	No realistic CF
STU-1348	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1349	0.03	0.03	0.03	0.05	Rejected	No realistic CF
STU-1350	0.04	0.02	0.02	0.01	Rejected	No realistic CF
STU-1351	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1352	0.06	0.07	0.09	0.09	Rejected	No realistic CF
STU-1353	0.03	0.06	0.06	0.06	Rejected	No realistic CF
STU-1354	0.07	0.06	0.09	0.05	Rejected	No realistic CF
STU-1355	0.06	0.08	0.09	0.07	Rejected	No realistic CF
STU-1356	0.02	0.04	0.06	0.04	Rejected	No realistic CF
STU-1357	0.46	0.40	0.62	0.55	Rejected	+0% M, +25% P, +20% C, +25% B
STU-1358	0.05	0.05	0.09	0.06	Rejected	No realistic CF
STU-1359	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1360	0.44	0.32	0.48	0.53	Rejected	+0% M, +35% P, +30% C, +25% B

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1361	0.05	0.09	0.09	0.09	Rejected	No realistic CF
STU-1362	0.06	0.06	0.05	0.04	Rejected	No realistic CF
STU-1363	0.06	0.06	0.09	0.07	Rejected	No realistic CF
STU-1364	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1365	0.03	0.01	0.04	0.04	Rejected	No realistic CF
STU-1366	0.08	0.08	0.09	0.04	Rejected	No realistic CF
STU-1367	0.80	0.64	0.78	0.75	Accepted	N/A (Accepted)
STU-1368	0.06	0.05	0.09	0.05	Rejected	No realistic CF
STU-1369	0.93	0.90	0.94	0.84	Accepted	N/A (Accepted)
STU-1370	0.09	0.07	0.09	0.06	Rejected	No realistic CF
STU-1371	0.01	0.01	0.01	0.00	Rejected	No realistic CF
STU-1372	0.38	0.31	0.48	0.25	Rejected	+30% M, +35% P, +40% C, +50% B
STU-1373	0.08	0.08	0.09	0.08	Rejected	No realistic CF
STU-1374	0.02	0.02	0.02	0.01	Rejected	No realistic CF
STU-1375	0.37	0.31	0.43	0.30	Rejected	+0% M, +35% P, +35% C, +50% B
STU-1376	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1377	0.04	0.02	0.03	0.03	Rejected	No realistic CF
STU-1378	0.05	0.05	0.06	0.06	Rejected	No realistic CF
STU-1379	0.08	0.05	0.05	0.05	Rejected	No realistic CF
STU-1380	0.76	0.47	0.67	0.63	Rejected	+0% M, +15% P, +10% C, +15% B
STU-1381	0.04	0.06	0.06	0.05	Rejected	No realistic CF
STU-1382	0.31	0.36	0.46	0.40	Rejected	+0% M, +30% P, +35% C, +40% B
STU-1383	0.02	0.05	0.06	0.04	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1384	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1385	0.04	0.02	0.04	0.03	Rejected	No realistic CF
STU-1386	0.01	0.01	0.01	0.02	Rejected	No realistic CF
STU-1387	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1388	0.28	0.20	0.44	0.17	Rejected	No realistic CF
STU-1389	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1390	0.03	0.04	0.04	0.05	Rejected	No realistic CF
STU-1391	0.02	0.02	0.01	0.01	Rejected	No realistic CF
STU-1392	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1393	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1394	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1395	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1396	0.06	0.09	0.09	0.09	Rejected	No realistic CF
STU-1397	0.06	0.09	0.07	0.05	Rejected	No realistic CF
STU-1398	0.04	0.08	0.09	0.09	Rejected	No realistic CF
STU-1399	0.03	0.02	0.02	0.02	Rejected	No realistic CF
STU-1400	0.04	0.08	0.09	0.09	Rejected	No realistic CF
STU-1401	0.01	0.02	0.02	0.01	Rejected	No realistic CF
STU-1402	0.69	0.60	0.71	0.56	Rejected	+10% M, +5% P, +10% C, +20% B
STU-1403	0.03	0.03	0.03	0.04	Rejected	No realistic CF
STU-1404	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1405	0.00	0.00	0.00	0.00	Rejected	No realistic CF
STU-1406	0.61	0.39	0.63	0.69	Rejected	+0% M, +25% P, +15% C, +10% B

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1407	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1408	0.05	0.07	0.09	0.09	Rejected	No realistic CF
STU-1409	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1410	0.02	0.02	0.02	0.02	Rejected	No realistic CF
STU-1411	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1412	0.63	0.54	0.65	0.62	Rejected	+0% M, +10% P, +15% C, +15% B
STU-1413	0.01	0.02	0.02	0.03	Rejected	No realistic CF
STU-1414	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1415	0.41	0.52	0.62	0.89	Rejected	+0% M, +15% P, +20% C, +0% B
STU-1416	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1417	0.07	0.06	0.06	0.05	Rejected	No realistic CF
STU-1418	0.05	0.04	0.08	0.04	Rejected	No realistic CF
STU-1419	0.06	0.09	0.07	0.08	Rejected	No realistic CF
STU-1420	0.36	0.24	0.40	0.34	Rejected	+0% M, +40% P, +40% C, +45% B
STU-1421	0.05	0.02	0.02	0.04	Rejected	No realistic CF
STU-1422	0.01	0.02	0.01	0.01	Rejected	No realistic CF
STU-1423	0.01	0.02	0.03	0.03	Rejected	No realistic CF
STU-1424	0.09	0.09	0.09	0.06	Rejected	No realistic CF
STU-1425	0.09	0.09	0.06	0.05	Rejected	No realistic CF
STU-1426	0.05	0.04	0.04	0.03	Rejected	No realistic CF
STU-1427	0.09	0.08	0.09	0.06	Rejected	No realistic CF
STU-1428	0.06	0.06	0.06	0.06	Rejected	No realistic CF
STU-1429	0.02	0.05	0.05	0.04	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1430	0.02	0.01	0.02	0.02	Rejected	No realistic CF
STU-1431	0.09	0.09	0.09	0.07	Rejected	No realistic CF
STU-1432	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1433	0.01	0.01	0.02	0.02	Rejected	No realistic CF
STU-1434	0.40	0.35	0.42	0.33	Rejected	+0% M, +30% P, +40% C, +45% B
STU-1435	0.06	0.05	0.06	0.06	Rejected	No realistic CF
STU-1436	0.03	0.02	0.02	0.01	Rejected	No realistic CF
STU-1437	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1438	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1439	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1440	0.42	0.28	0.31	0.31	Rejected	+0% M, +40% P, +50% C, +50% B
STU-1441	0.27	0.20	0.39	0.30	Rejected	+0% M, +45% P, +40% C, +50% B
STU-1442	0.09	0.09	0.09	0.08	Rejected	No realistic CF
STU-1443	0.40	0.23	0.46	0.22	Rejected	No realistic CF
STU-1444	0.43	0.26	0.45	0.38	Rejected	+0% M, +40% P, +35% C, +40% B
STU-1445	0.71	0.65	0.76	0.74	Rejected	+0% M, +0% P, +5% C, +5% B
STU-1446	0.85	0.78	0.95	0.81	Accepted	N/A (Accepted)
STU-1447	0.05	0.04	0.04	0.02	Rejected	No realistic CF
STU-1448	0.30	0.22	0.42	0.31	Rejected	+0% M, +40% P, +45% C, +50% B
STU-1449	0.03	0.05	0.08	0.05	Rejected	No realistic CF
STU-1450	0.04	0.07	0.05	0.05	Rejected	No realistic CF
STU-1451	0.03	0.01	0.01	0.01	Rejected	No realistic CF
STU-1452	0.03	0.02	0.02	0.01	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1453	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1454	0.09	0.09	0.09	0.08	Rejected	No realistic CF
STU-1455	0.40	0.24	0.36	0.33	Rejected	+0% M, +40% P, +45% C, +45% B
STU-1456	0.02	0.01	0.01	0.01	Rejected	No realistic CF
STU-1457	0.02	0.02	0.02	0.04	Rejected	No realistic CF
STU-1458	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1459	0.05	0.07	0.06	0.05	Rejected	No realistic CF
STU-1460	0.01	0.00	0.00	0.01	Rejected	No realistic CF
STU-1461	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1462	0.04	0.03	0.02	0.04	Rejected	No realistic CF
STU-1463	0.02	0.02	0.01	0.01	Rejected	No realistic CF
STU-1464	0.87	0.58	0.78	0.72	Rejected	+0% M, +5% P, +0% C, +5% B
STU-1465	0.73	0.74	0.90	0.54	Rejected	+0% M, +0% P, +0% C, +25% B
STU-1466	0.39	0.33	0.57	0.36	Rejected	+0% M, +30% P, +30% C, +45% B
STU-1467	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1468	0.52	0.41	0.67	0.39	Rejected	+0% M, +25% P, +15% C, +40% B
STU-1469	0.07	0.09	0.09	0.06	Rejected	No realistic CF
STU-1470	0.01	0.01	0.01	0.01	Rejected	No realistic CF
STU-1471	0.07	0.09	0.09	0.07	Rejected	No realistic CF
STU-1472	0.49	0.58	0.61	0.74	Rejected	+0% M, +10% P, +20% C, +5% B
STU-1473	0.09	0.08	0.09	0.05	Rejected	No realistic CF
STU-1474	0.01	0.02	0.03	0.03	Rejected	No realistic CF
STU-1475	0.05	0.09	0.05	0.05	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	Math	Phys	Chem	Bio	Current Status	Actionable Advice (△)
STU-1476	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1477	0.03	0.03	0.02	0.04	Rejected	No realistic CF
STU-1478	0.05	0.02	0.03	0.04	Rejected	No realistic CF
STU-1479	0.02	0.03	0.04	0.04	Rejected	No realistic CF
STU-1480	0.01	0.04	0.03	0.03	Rejected	No realistic CF
STU-1481	0.00	0.00	0.00	0.01	Rejected	No realistic CF
STU-1482	0.04	0.03	0.03	0.03	Rejected	No realistic CF
STU-1483	0.09	0.09	0.06	0.06	Rejected	No realistic CF
STU-1484	0.03	0.02	0.02	0.02	Rejected	No realistic CF
STU-1485	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1486	0.08	0.09	0.09	0.09	Rejected	No realistic CF
STU-1487	0.04	0.06	0.07	0.05	Rejected	No realistic CF
STU-1488	0.51	0.38	0.41	0.54	Rejected	+15% M, +25% P, +40% C, +25% B
STU-1489	0.05	0.05	0.09	0.04	Rejected	No realistic CF
STU-1490	0.03	0.06	0.08	0.05	Rejected	No realistic CF
STU-1491	0.01	0.01	0.01	0.00	Rejected	No realistic CF
STU-1492	0.06	0.09	0.09	0.07	Rejected	No realistic CF
STU-1493	0.09	0.09	0.06	0.05	Rejected	No realistic CF
STU-1494	0.03	0.05	0.07	0.05	Rejected	No realistic CF
STU-1495	0.01	0.00	0.01	0.01	Rejected	No realistic CF
STU-1496	0.04	0.09	0.09	0.09	Rejected	No realistic CF
STU-1497	0.09	0.09	0.09	0.09	Rejected	No realistic CF
STU-1498	0.08	0.08	0.09	0.06	Rejected	No realistic CF

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Table 32 – continued from previous page

Student ID	**Math**	**Phys**	**Chem**	**Bio**	**Current Status**	**Actionable Advice (Δ)**
STU-1499	0.49	0.37	0.57	0.36	Rejected	+20% M, +25% P, +30% C, +40% B

Appendix D: Request for Data Access

The following is the original request letter submitted to the Ministry of Education for data access to support this research.

MURENZI Benjamin

Master's Student, Big Data Analytics
Adventist University of Central Africa(AUCA)
Tel. : 0788560661
Email: ngingaben1@gmail.com

Pour réception
Secrétariat Central / MINEDUC
le 20/06/2024 Signature

To: Permanent Secretary
Ministry of Education
KIGALI, RWANDA

ADVENTIST UNIVERSITY OF CENTRAL AFRICA RWANDA

Through: Dr. Pacifique NIZEYIMANA
Supervisor and Big Data Analytics Program Coordinator

Dear PS,

Subject: Request for Data Access for Research on Career Path Guidance Model with Multi-Label Classification

I am writing this letter to request data access for my academic research project titled "Career Path Guidance Model with Multi-Label Classification," under the supervision of Dr. Pacifique NIZEYIMANA and Dr. Kumar Kundan.

I am requesting data on: Academic performances, graduation statistics and any other relevant metrics that can contribute to the classification process. I assure that the data will be handled with confidentiality and used for academic purposes. This research has the potential to significantly benefit the educational sector.

Thank you for considering my request; I look forward to your positive response.

Yours sincerely,

19th/06/2024

MURENZI Benjamin

Cc:

- Director General, Higher Education Council (HEC)
- Director General, National Examination and School Inspection Authority (NESA)
- Director General, Rwanda Education Board (REB)
- Director General, Rwanda TVET Board (RTB)

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